

The Nervous System: A. General Principles and Sensory Physiology

- Which ion has the greatest electrochemical driving force in a typical neuron with a resting membrane potential of -65 millivolts?
 - Chloride
 - Potassium
 - Sodium
- A 2-year-old girl with fever is hyperventilating. Which of the following is most likely to occur in this girl?
 - Decreased brain oxygenation only
 - Decreased brain oxygenation and increased neuronal activity
 - Decreased neuronal activity only
 - Increased brain oxygenation only
 - Increased brain oxygenation and decreased neuronal activity
 - Increased neuronal activity only
- Pain receptors in the skin are typically classified as which of the following?
 - Encapsulated nerve endings
 - A single class of morphologically specialized receptors
 - The same type of receptor that detects position sense
 - Free nerve endings
- Which of the following best describes an expanded tip tactile receptor found in the dermis of hairy skin that is specialized to detect continuously applied touch sensation?
 - Free nerve endings
 - Merkel disc
 - Pacinian corpuscle
 - Ruffini endings
- The release of neurotransmitter at a chemical synapse in the central nervous system is dependent upon which of the following?
 - Synthesis of acetylcholinesterase
 - Hyperpolarization of the synaptic terminal
 - Opening of ligand-gated ion calcium channels
 - Influx of calcium into the presynaptic terminal
- Which of the following is best described as an elongated, encapsulated receptor found in the dermal peps of glabrous skin that is especially abundant on lips and fingertips?
 - Merkel disc
 - Free nerve endings
 - Meissner corpuscle
 - Ruffini endings
- A transmitter substance released from a presynaptic neuron activates a second messenger G-protein system in the postsynaptic neuron. Which one of the following postsynaptic responses to the transmitter substance is NOT a possible outcome?
 - Activation of cyclic adenosine monophosphate (cAMP)
 - Activation of cyclic guanosine monophosphate (cGMP)
 - Activation of gene transcription
 - Closing an ion channel
 - Opening an ion channel
- A 43-year-old man sustained a lower back injury that causes severe chronic pain. His physician prescribes benzodiazepine sedation medications to help him sleep. Which response best describes why this man has difficulty sleeping without medication?
 - Depression of the amygdala
 - Depression of reticular formation
 - Excitation of the amygdala
 - Excitation of reticular formation
 - Loss of somatic sensations
 - Loss of visceral sensations
- A 15-year-old girl with epilepsy visits a physician for testing. The physician uses electroencephalography to study her brain waves during various activities. Which of the following is most likely to stimulate the greatest increase in brain activity in this girl?
 - Hyperventilation
 - Hypoventilation
 - Hyperventilation plus flashing lights
 - Hypoventilation plus flashing lights

10. Which of the following best describes the concept of specificity in sensory nerve fibers that transmit only one modality of sensation?
 - A) Frequency coding principle
 - B) Concept of specific nerve energy
 - C) Singularity principle
 - D) Labeled line principle
11. Which of the following is an encapsulated receptor found deep in the skin throughout the body, as well as in fascial layers, where it detects indentation of the skin (pressure) and movement across the surface (vibration)?
 - A) Pacinian corpuscle
 - B) Meissner's corpuscle
 - C) Free nerve endings
 - D) Ruffini endings
12. Which substance enhances the sensitivity of pain receptors but does not directly excite them?
 - A) Bradykinin
 - B) Serotonin
 - C) Potassium ions
 - D) Prostaglandins
13. Which of the following is an important functional parameter of pain receptors?
 - A) Exhibit little or no adaptation
 - B) Not affected by muscle tension
 - C) Signal only flexion at joint capsules
 - D) Can voluntarily be inhibited
14. The excitatory or inhibitory action of a neurotransmitter is determined by which of the following?
 - A) The function of its postsynaptic receptor
 - B) Its molecular composition
 - C) The shape of the synaptic vesicle in which it is contained
 - D) The distance between the pre- and postsynaptic membranes
15. A 39-year-old neurosurgeon picks up a scalpel, which activates numerous sensory receptors in her hand. An increase in which of the following best describes the basis for transduction of the sensory stimuli into nerve impulses?
 - A) Activation of G protein
 - B) Decreased ion permeability
 - C) Decreased transmitter release
 - D) Increased ion permeability
 - E) Increased transmitter release
 - F) Inhibition of G protein
16. Which ion has the lowest electrochemical driving force in a typical neuron with a resting membrane potential of -65 millivolts?
 - A) Chloride
 - B) Potassium
 - C) Sodium
17. A physiology experiment is conducted in which a glass microelectrode is inserted into a Pacinian corpuscle to record receptor potentials during different levels of stimulation (from 0 percent to 100 percent). Increasing stimulus strength from 10 percent of maximum to 30 percent of maximum causes a 40 percent increase in the amplitude of the receptor potential. Increasing the stimulus potential from 70 percent of maximum to 90 percent of maximum is most likely to cause which increase in the amplitude of the receptor potential (in percent)?
 - A) 10
 - B) 40
 - C) 60
 - D) 80
18. Interneurons that utilize the neurotransmitter enkephalin to inhibit afferent pain signals are most likely to be found in which region of the central nervous system?
 - A) Dorsal horn of spinal cord
 - B) Postcentral gyrus
 - C) Precentral gyrus
 - D) δ -type A
 - E) Type C fiber
 - F) Ventral horn of spinal cord
19. Which system transmits somatosensory information with the highest degree of temporal and spatial fidelity?
 - A) Anterolateral system
 - B) Dorsal column–medial lemniscal system
 - C) Corticospinal system
 - D) Spinocerebellar system
20. The pathway of which system crosses in the ventral white commissure of the spinal cord within a few segments of entry and then courses to the thalamus contralateral to the side of the body from which the signal originated?
 - A) Anterolateral system
 - B) Dorsal column–medial lemniscal system
 - C) Corticospinal system
 - D) Spinocerebellar system
21. Neurons located in which area release serotonin as their neurotransmitter?
 - A) Periaqueductal gray area
 - B) Interneurons of the spinal cord
 - C) Periventricular area
 - D) Nucleus raphe magnus
22. Which system conveys information concerning highly localized touch sensation and body position (proprioceptive) sensation?
 - A) Anterolateral
 - B) Dorsal column–medial lemniscal
 - C) Corticospinal
 - D) Spinocerebellar

23. The first-order (primary afferent) cell bodies of the dorsal column–medial lemniscal system are found in which structure?
- Spinal cord dorsal horn
 - Spinal cord ventral horn
 - Dorsal root ganglia**
 - Nucleus cuneatus
24. Which structure carries axons from the nucleus gracilis to the thalamus?
- Fasciculus gracilis
 - Fasciculus lemniscus
 - Lateral spinothalamic tract
 - Medial lemniscus**
25. A 10-year-old boy cuts his finger with a pocketknife and immediately applies pressure to the damaged area with his other hand to partially alleviate the pain. Inhibition of pain signals by tactile stimulation of the skin is mediated by which type of afferent neurons from mechanoreceptors?
- α -type A
 - β -type A**
 - δ -type A
 - Type C
26. A pool of presynaptic neurons innervate the dendrites of a postsynaptic neuron. Electrical signals are transferred from the dendrites to the soma of the postsynaptic neuron by which process?
- Action potential
 - Active transport
 - Capacitive discharge
 - Diffusion
 - Electrotonic conduction**
27. Which structure carries axons from neurons in the ventral posterolateral nucleus of the thalamus to the primary somatosensory cortex?
- Medial lemniscus
 - External capsule
 - Internal capsule**
 - Extreme capsule
28. Which of the following is characteristic of the events occurring at an excitatory synapse?
- There is a massive efflux of calcium from the presynaptic terminal
 - Synaptic vesicles bind to the postsynaptic membrane
 - Voltage-gated potassium channels are closed
 - Ligand-gated channels are opened to allow sodium entry into the postsynaptic neuron**
29. Stimulation of which brain area can modulate the sensation of pain?
- Superior olivary complex
 - Locus ceruleus
 - Periaqueductal gray area**
 - Amygdala
30. Which body part is represented superiorly and medially within the postcentral gyrus?
- Upper limb
 - Lower limb**
 - Abdomen
 - Genitalia
31. Which of the following is a group of neurons in the pain suppression pathway that uses enkephalin as a neurotransmitter?
- Postcentral gyrus
 - Nucleus raphe magnus
 - Periaqueductal gray area**
 - Type AB sensory fibers
- Questions 32 and 33**
- A 19-year-old man has an automobile accident that completely eliminates all nerve traffic in the right half of the spinal cord at C2. Use this information to answer Questions 32 and 33.
32. Loss of which function is most likely in the right hand of this man?
- Crude touch and pain sensation
 - Crude touch and temperature sensation
 - Motor function and temperature sensation**
 - Motor function and vibration sense**
 - Vibration sense and crude touch
 - Vibration sense and pain sensation
33. Loss of which function is most likely in the left hand of this man?
- Crude touch and pain sensation**
 - Crude touch and vibration sense
 - Motor function and temperature sensation
 - Motor function and vibration sense
 - Vibration sense and pain sensation
 - Vibration sense and crude touch
34. The highest degree of pain localization comes from which of the following?
- Simultaneous stimulation of free nerve endings and tactile fibers**
 - Stimulation of free nerve endings by bradykinin
 - Nerve fibers traveling to the thalamus by way of the paleospinothalamic tract
 - Stimulation of δ -type A fibers

35. Which of the following is the basis for referred pain?
- Visceral pain signals and pain signals from the skin synapse with separate populations of neurons in the dorsal horn
 - Visceral pain transmission and pain transmission from the skin are received by a common set of neurons in the thalamus
 - Visceral pain signals are rarely of sufficient magnitude to exceed the threshold of activation of dorsal horn neurons
 - Some visceral pain signals and pain signals from the skin provide convergent input to a common set of neurons in the dorsal horn
36. Post-tetanic facilitation is thought to be the result of which of the following?
- Opening of voltage-gated sodium channels
 - Opening of transmitter-gated potassium channels
 - A buildup of calcium in the presynaptic terminal
 - Electrotonic conduction
37. Within the primary somatosensory cortex, the various parts of the contralateral body surface are represented in areas of varying size that reflect which of the following?
- The relative size of the body parts
 - The density of the specialized peripheral receptors
 - The size of the muscles in that body part
 - The conduction velocity of the primary afferent fibers
38. The gray matter of the primary somatosensory cortex contains six layers of cells. Which layer(s) receive the bulk of incoming signals from the somatosensory nuclei of the thalamus?
- I
 - II and III
 - III only
 - IV
39. Which statement concerning visceral pain signals is correct?
- They are transmitted along sensory fibers that course mainly with sympathetic nerves in the abdomen and thorax
 - They are not stimulated by ischemia in visceral organs
 - They are transmitted only by the lightly myelinated δ -type A sensory fibers
 - They are typically well localized
40. Pain from the stomach is referred to which area of the body?
- Upper right shoulder area
 - Abdominal area above the umbilicus
 - Proximal area of the anterior and inner thigh
 - Abdominal area below the umbilicus

Questions 41–43

Each of the disorders in Questions 41–43 is characterized either by the production of excessive pain (hyperalgesia) or the loss of pain sensation.

41. Which disorder is characterized by excessive pain in a skin dermatomal distribution resulting from a viral infection of a dorsal root ganglion?
- Tic douloureux
 - Thalamic pain syndrome
 - Brown-Séquard syndrome
 - Herpes zoster
42. Which disorder involves a loss of pain sensation on one side of the body coupled with the loss of proprioception, precise tactile localization, and vibratory sensations on the contralateral side of the body?
- Herpes zoster
 - Thalamic pain syndrome
 - Lateral medullary syndrome
 - Brown-Séquard syndrome
43. Which disorder is characterized by the loss of pain sensation throughout one entire side of the body and the opposite side of the face?
- Brown-Séquard syndrome
 - Thalamic pain syndrome
 - Herpes zoster
 - Lateral medullary syndrome
44. Stimulation by touching or pulling on which structure is least likely to cause a painful sensation?
- The postcentral gyrus
 - The dura overlying the postcentral gyrus
 - Branches of the middle meningeal artery that lie superficial to the dura over the postcentral gyrus
 - Branches of the middle cerebral artery that supply the postcentral gyrus
45. Vibratory sensation depends on the detection of rapidly changing, repetitive sensations. The high-frequency end of the repetitive stimulation scale is detected by which structure?
- Merkel discs
 - Meissner corpuscles
 - Pacinian corpuscles
 - Free nerve endings
46. The ability to detect two points simultaneously applied to the skin is based on which physiologic mechanism?
- Presynaptic inhibition
 - Lateral inhibition
 - Medial inhibition
 - Feed-forward inhibition

47. Which electrical event is characteristic of inhibitory synaptic interactions?
- A neurotransmitter agent that selectively opens ligand-gated chloride channels is the basis for an inhibitory postsynaptic potential
 - Because the Nernst potential for chloride is about -70 mV, chloride ions tend to move out of the cell along its electrochemical gradient
 - A neurotransmitter that selectively opens potassium channels will allow potassium to move into the cell
 - An increase in the extracellular sodium concentration usually leads directly to an inhibitory postsynaptic potential
48. Which somatosensory deficit is NOT typically seen after the development of lesions that involve the post-central gyrus?
- Inability to discretely localize touch sensation over the contralateral face and upper limb
 - Inability to judge the weight of easily recognizable objects
 - Inability to accurately assess the texture of common objects by touching them with the fingers
 - Inability to move the contralateral arm and leg
49. Which statement concerning sensory neurons or their functional properties is true?
- All sensory fibers are unmyelinated
 - In spatial summation, increasing signal strength is transmitted by using progressively greater numbers of sensory fibers
 - Increased stimulus intensity is signaled by a progressive decrease in the receptor potential
 - Continuous subthreshold stimulation of a pool of sensory neurons results in disfacilitation of those neurons
 - Temporal summation involves signaling of increased stimulus strength by decreasing the frequency of action potentials in the sensory fibers
50. A 23-year-old gymnast lifts her right leg above her head while in the standing position. Activation of a single pyramidal cell in the motor cortex leads to stimulation of 2000 muscle fibers in her right quadriceps muscle. Which of the following best describes the type of neuronal circuitry activated in this woman when she lifts her leg?
- Converging
 - Diverging
 - Inhibitory
 - Reverberatory
51. An input neuron to a diverging circuit causes the membrane potential of a target neuron to change from -65 millivolts to -55 millivolts. Which of the following best describes this change in membrane potential (in millivolts)?
- Excitatory postsynaptic potential = $+10$
 - Excitatory postsynaptic potential = -10
 - Inhibitory postsynaptic potential = $+10$
 - Inhibitory postsynaptic potential = -10
52. Prolonged changes in neuronal activity are usually achieved through the activation of which of the following?
- Voltage-gated chloride channels
 - Transmitter-gated sodium channels
 - G-protein-coupled channels
 - Voltage-gated potassium channels
53. Position sense, or more commonly proprioceptive sensation, involves muscle spindles and which of the following?
- Skin tactile receptors
 - Deep receptors in joint capsules
 - Both tactile and joint capsule receptors
 - Pacinian corpuscles
54. Migraine headaches often begin with a prodromal symptom such as nausea, loss of vision, visual aura, or other sensory hallucinations. Which of the following is believed to be the cause of such prodromes?
- Increased blood flow to brain tissue in the visual or other sensory cortex
 - A selective loss of gamma-aminobutyric acid neurons in the various sensory areas of cortex
 - Constipation
 - Vasospasm leading to ischemia and a disruption of neuronal activity in the relevant sensory areas of cortex
55. Which statement accurately describes a feature of temperature sensation by the nervous system?
- Cold receptors continue to be activated even if skin temperature is lowered well below its freezing point
 - Both cold and warm receptors each have very specific, nonoverlapping ranges of temperature sensitivity
 - Warm and cold receptors respond to both steady state temperatures and to changes in temperature
 - Temperature receptor function is the result of ion conduction changes and not changes in their metabolic rate
56. For a sensory nerve fiber that is connected to a Pacinian corpuscle located on palmar surface of the right hand, the synaptic connection with the subsequent neuron in the corresponding sensory pathway is located in
- The right dorsal column nucleus
 - The left dorsal column nucleus
 - The dorsal horn of the right side of the spinal cord
 - The dorsal horn of the left side of the spinal cord

57. The sensation of temperature is signaled mainly by warm and cold receptors whose sensory fibers travel in association with the sensory fibers carrying pain signals. Which statement best characterizes the transmission of signals from warm receptors?
- A) Warm receptors are well characterized histologically
 - B) Signals from warm receptors are mainly transmitted along slow-conducting type C sensory fibers
 - C) Warm receptors are located well below the surface of the skin in the subcutaneous connective tissue
 - D) There are 3 to 10 times more warm receptors than cold receptors in most areas of the body
58. Like other sensory systems, the somatosensory system has a descending component that functions to regulate the overall sensitivity of the system. Which of the following selections best describes the function of the corticofugal signals transmitted from the somatosensory cortex downward to the thalamus and dorsal column nuclei?
- A) Increase or decrease the perception of signal intensity
 - B) Decrease the ability to detect body position sense
 - C) Remove the thalamus from the processing of somatosensory signals
 - D) Allow ascending information to bypass the nucleus cuneatus and gracilis
59. Which statement concerning the generation of an action potential is correct?
- A) When the membrane potential in the soma/axon hillock dips below "threshold," an action potential is initiated
 - B) The action potential is initiated in synaptic boutons
 - C) The least number of voltage-gated sodium channels in an axon is found near the node of Ranvier
 - D) Once an action potential is initiated, it will always run its course to completion
60. Which statement concerning synaptic transmission is correct?
- A) When a specific population of synaptic terminals is spread over the considerable surface of a neuron, their collective effects cannot spatially summate and lead to initiation of an action potential
 - B) Even if the successive discharges of an excitatory synapse occur sufficiently close in time, they cannot temporally summate and initiate an action potential
 - C) A neuron is "facilitated" when its membrane potential is moved in the less negative or depolarizing direction
 - D) Even when rapidly stimulated by excitatory synaptic input for a prolonged period, neurons typically do not exhibit synaptic fatigue

ANSWERS

1. **C)** The electrochemical driving force (V_{DF}) for an ion can be calculated as follows: $V_{DF} = V_m - V_{eq}$, where V_m is the membrane potential and V_{eq} is the equilibrium potential of the ion. A calculator is available at http://physiology-web.com/calculators/electrochemical_driving_force_calculator.html. A positive value indicates outward flux of the ion, and a negative value indicates inward flux of the ion. A typical equilibrium potential for sodium (calculated using the Nernst equation) is +62 millivolts, so the electrochemical driving force for sodium is $-65 - 62 = -127$ millivolts. This means that a 127-millivolt force attempts to drive sodium into the cell. The equilibrium potential is about -86 millivolts for potassium and about -70 millivolts for chloride; hence, the electrochemical driving force for these two ions is +21 and +5 millivolts, respectively (and both ions tend to be driven out of the cell).
TMP13 pp. 587-588
2. **B)** Hyperventilation lowers the carbon dioxide tension of the blood, which leads to increases in the pH of the body tissues, including the brain. Alkalinity increases neuronal activity in the brain. Carbon dioxide also has the potent effect of increasing cerebral blood flow; thus, hyperventilation can lead to decreased cerebral blood flow with a subsequent decrease in oxygenation of the brain.
TMP13 pp. 592, 787
3. **D)** Pain receptors in the skin are free nerve endings.
TMP13 p. 621
4. **B)** Merkel discs are found in the dermis of hairy skin and signal continuous touch.
TMP13 p. 608
5. **D)** The release of neurotransmitter depends on the influx of calcium through voltage-gated channels. When this influx occurs, synaptic vesicles fuse with the pre-synaptic membrane and release the transmitter agent into the synaptic cleft.
TMP13 p. 582
6. **C)** Meissner corpuscles are found in the dermal pegs.
TMP13 pp. 596, 607-608
7. **D)** A so-called second messenger system can be activated by a transmitter substance released from an initial neuron by first causing the release of a G protein into the second neuron's cytoplasm. Neurotransmitter activation of G proteins is not known to cause closure of an ion channel. G proteins can activate G-protein-gated ion channels for both sodium and potassium, as well as gene transcription, and cAMP and cGMP. G proteins also can activate intracellular enzymes that have a variety of different functions.
TMP13 pp. 583-584
8. **D)** Individuals experiencing severe chronic pain have difficulty sleeping because the ascending pain pathways provide excitatory input to reticular formation elements that constitute the reticular activating system; this system maintains the alert, waking state. The overall function of the amygdala is thought to make the person's behavioral response appropriate for each occasion; it does not play a major role in establishing the awake state. Loss of visceral sensations or somatic sensations would likely help the man sleep.
TMP13 pp. 623-624
9. **C)** Hyperventilation plus flashing lights can sometimes initiate an epileptic seizure in a susceptible person who is poorly medicated. Flashing lights alone activate neurons in the occipital cortex that can sometimes lead to increases in electrical activity throughout the brain. Hyperventilation (taking long, deep breaths) lowers carbon dioxide levels in the blood, causing the brain to become alkalotic; this activation method is commonly used to increase brain activity during electroencephalography.
TMP13 p. 592
10. **D)** The association of one sensory modality with one type of nerve fiber is the basis for the labeled line theory.
TMP13 p. 595
11. **A)** Pacinian corpuscles detect pressure and movement across the skin surface and are encapsulated receptors found deep in the skin throughout the body.
TMP13 pp. 597, 608
12. **D)** Prostaglandins are believed to enhance the sensitivity of pain receptors but do not actually excite them.
TMP13 p. 621
13. **A)** Pain receptors exhibit little or no functional adaptation.
TMP13 pp. 621-622
14. **A)** The function of a transmitter agent is solely dependent on the postsynaptic receptor to which it binds.
TMP13 p. 582
15. **D)** Virtually all mechanical stimuli cause an increase in ion permeability (usually to sodium) in mechanoreceptors. If the membrane potential of the mechanoreceptor reaches a critical threshold value, an action potential is initiated. The G-protein "second messenger" system is typically involved with prolonged postsynaptic neuronal excitation or inhibition; transduction in mechanoreceptors is rapid and transient. Transmitter release does

not occur at the level of the mechanoreceptor, but if a mechanoreceptor is activated, afferent nerve impulses do stimulate transmitter release at the nerve terminal in the central nervous system.

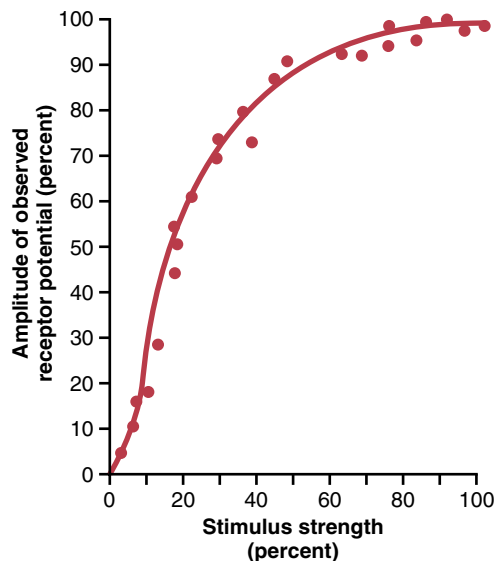
TMP13 pp. 582-583

- 16. A)** The equilibrium potential for chloride averages about -70 millivolts in a typical neuron, so with a resting membrane potential of -65 millivolts, the electrochemical driving force for chloride is 5 millivolts. The electrochemical driving force for sodium is about 127 millivolts; potassium has a value of about 21 millivolts during basal conditions.

TMP13 pp. 587-588

- 17. A)** The amplitude of the receptor potential from a Pacinian corpuscle increases greatly with a step increase in stimulus intensity at lower levels of stimulus strength, and to a lesser extent with a similar step increase at higher levels of stimulus strength, as shown in the figure below. This relationship between stimulus strength and amplitude of receptor potential allows the Pacinian corpuscle to discern small changes in stimulus strength at low levels of stimulation and yet still respond to changes in stimulus strength when the intensity of stimulation is high.

TMP13 pp. 597-598



Data from Loewenstein WR: *Excitation and inactivation in a receptor membrane*. *Ann N Y Acad Sci* 94:510, 1961.

- 18. A)** Interneurons in the dorsal horn of the spinal cord use enkephalin as a transmitter substance that effectively inhibits pain transmission from tissues of the body. The somatosensory cortex is located in the post-central gyrus, and the primary motor cortex is located in the precentral gyrus; neither are thought to use enkephalin to inhibit pain transmission. Myelinated δ -type A fibers and unmyelinated type C fibers are not interneurons. Interneurons are physically short neurons that form a connection between other neurons that are usually close together. There are distinguished from "projection" neurons that project to more distant regions of the brain or spinal cord.

TMP13 pp. 625-626

- 19. B)** Temporal and spatial fidelity is enhanced in the dorsal column–medial lemniscal system compared with the anterolateral system.

TMP13 p. 610

- 20. A)** Fibers in the anterolateral system cross in the anterior white commissure within a few segments of their entry before ascending on the contralateral side. Signals ascending in the dorsal column–medial lemniscal system do not cross until they reach the dorsal column nuclei in the medulla.

TMP13 pp. 616-617

- 21. D)** Neurons of the nucleus raphe magnus release serotonin at their nerve endings. In the endogenous pain suppression system, the termination of these neurons is in the spinal cord on interneurons that in turn release enkephalin and block the incoming signals from the pain fibers.

TMP13 p. 625

- 22. B)** The sensations of highly localized touch and body position are carried in the dorsal column–medial lemniscal system.

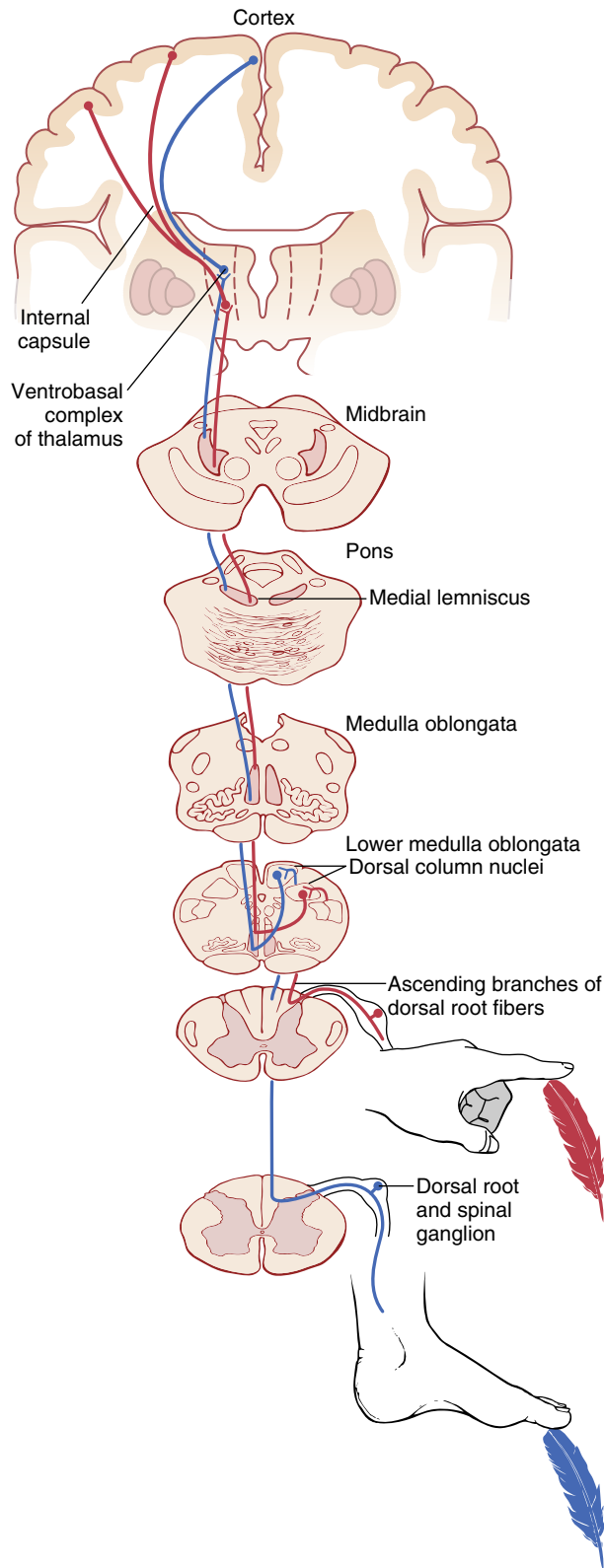
TMP13 p. 610

- 23. C)** Primary afferent neuronal cell bodies are found in the dorsal root ganglia.

TMP13 pp. 609-610

- 24. D)** The medial lemniscus conveys axons from the nucleus gracilis and cuneatus to the thalamus (see the figure below).

TMP13 p. 610



- 25. B)** Stimulation of β -type A fibers from peripheral tactile receptors can decrease transmission of pain signals by a type of lateral inhibition; this process is mediated by inhibitory interneurons in the dorsal column of the spinal cord. α -Type A neurons project to skeletal muscles, causing them to contract. δ -Type A fibers and type C fibers conduct pain signals to the dorsal column of the spinal cord.

TMP13 pp. 599-600

- 26. E)** Transmission of electrical signals in dendrites occurs by electrotonic conduction. Dendrites have few voltage-gated sodium channels, which makes it virtually impossible for action potentials to be initiated in this portion of a typical neuron. A neuron can be considered as a type of capacitor that discharges during an action potential, but this occurs in the axon, not the dendrites. Electrotonic conduction does not occur by diffusion or active transport.

TMP13 pp. 590-591

- 27. C)** The internal capsule conveys axons from the ventral posterolateral thalamic nucleus to the primary somatosensory cortex.

TMP13 p. 610

- 28. D)** Ligand-gated channels open and allow entry of sodium. This entry is accompanied by the influx of calcium, binding of synaptic vesicles to the presynaptic membrane, and electrical changes in the postsynaptic membrane.

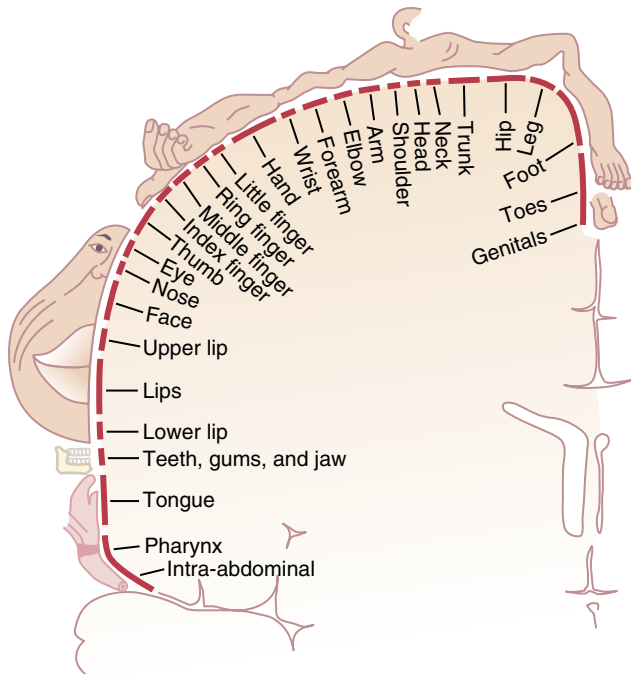
TMP13 pp. 582-583

- 29. C)** The periaqueductal gray area in the midbrain contains neurons that contribute to the descending pain suppression system.

TMP13 p. 625

- 30. B)** The lower limb representation is found in the superior and medial portion of the postcentral gyrus (see the figure below).

TMP13 p. 612



From Penfield W, Rasmussen T: *The Cerebral Cortex of Man*. Copyright 1950 Gale, a part of Cengage Learning, Inc. Reproduced by permission. www.cengage.com/permissions.

- 31. C)** Neurons in the periaqueductal gray area use enkephalin as a transmitter agent.

TMP13 p. 625

- 32. D)** Most motor neurons cross to the contralateral side in the pyramidal decussation of the medulla oblongata, which is proximal to the damaged area. Fine sensory sensations (vibration sense, fine touch, proprioception, and two-point discrimination) transmitted in the dorsal-column medial lemniscal pathway cross to the contralateral side in the medulla. Therefore, both motor function and vibration sense are lost on the same side (ipsilateral) as the cord lesion.

TMP13 pp. 610, 709

- 33. A)** Crude touch, pain sensations, and temperature sensations travel in the anterolateral pathway of the spinal cord; the afferent neurons from the receptor organs decussate in the spinal cord close to the point of entry. Hence, these sensations are lost on the side opposite of the lesion.

TMP13 pp. 609-610

- 34. A)** In general, the sensation of pain is poorly localized. However, when a tactile receptor and a pain receptor are stimulated simultaneously, the pain sensation is localized with greater accuracy.

TMP13 p. 624

- 35. D)** Visceral pain fibers can provide input to anterolateral tract cells that also receive somatic pain from the skin surface. The convergence of these two types of pain signals onto single spinal cord neurons is thought to be the basis for referred pain.

TMP13 p. 626

- 36. C)** Post-tetanic facilitation is the neuronal phenomenon in which a neuron is more easily excited after a brief period of activity. This phenomenon is thought to be due to the buildup of calcium in the presynaptic membrane caused by the prior neuronal activity. Subsequent neuronal impulses release neurotransmitter more readily as a result of this preplaced calcium from the prior stimulus.

TMP13 pp. 589-590

- 37. B)** The size of the representation of various body parts in the primary somatosensory cortex is correlated with the density of cutaneous receptors in that body part.

TMP13 p. 612

- 38. D)** Layer IV of the somatosensory cortex receives the bulk of the input from the somatosensory nuclei of the thalamus.

TMP13 p. 612

- 39. A)** Visceral pain signals from structures in the abdomen and thorax travel toward the spinal cord in association with fibers of the sympathetic system.

TMP13 pp. 627-628

- 40. B)** Pain from the stomach is referred to the upper abdominal area. In general, it will be above the level of the umbilicus.

TMP13 p. 627

- 41. D)** Herpes zoster is a disorder characterized by excessive pain in a dermatomal distribution that results from a viral infection of a dorsal root ganglion.

TMP13 p. 628

- 42. D)** The Brown-Séquard syndrome is characterized by the loss of pain sensation on one side of the body coupled with a loss of discriminative sensations, such as proprioception and vibratory sensation, on the opposite side of the body.

TMP13 p. 628

- 43. D)** The lateral medullary syndrome exhibits one of the most characteristic patterns of sensory loss in clinical neurology; pain sensation is lost over one side of the body from feet to neck and on the opposite side of the face. Moreover, the side of facial pain loss indicates the side of the lesion.

TMP13 p. 628

- 44. A)** Touching or pulling on the postcentral gyrus is least likely to evoke a painful sensation because brain tissue lacks pain receptors.

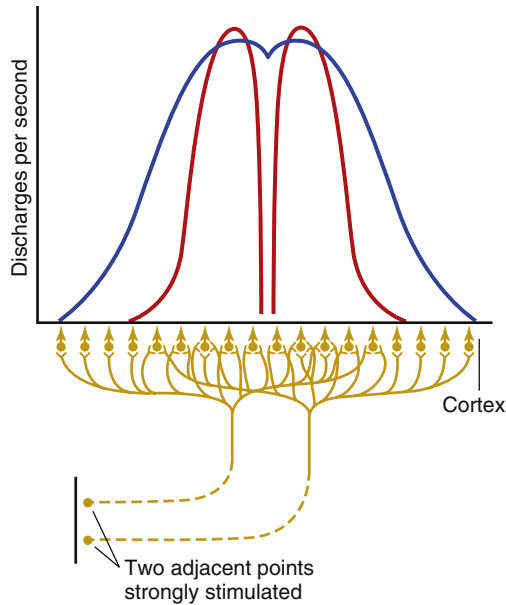
TMP13 p. 629

45. C) High-frequency repetitive stimulation (indentation/pressure) of the skin is sensed by Pacinian corpuscles.

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46. B) The process of lateral inhibition, illustrated in the figure below, underlies the ability to discriminate two points simultaneously applied.

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47. A) The opening of ligand-gated chloride channels and movement of chloride ions into the cell leads to hyperpolarization of the membrane. Neither increasing the extracellular sodium concentration nor the movement of potassium into the cell leads to hyperpolarization of the membrane.

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48. D) Paralysis of the contralateral arm and leg is a motor deficit, and such a deficit would not typically be observed after damage to the primary somatosensory cortex.

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49. B) In spatial summation, increasing signal strength is transmitted by using greater numbers of sensory fibers.

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50. B) A diverging neuronal pathway amplifies nerve signals; activation of a single pyramidal cell in motor cortex can stimulate as many as 10,000 muscle fibers. In a converging circuit, multiple input fibers converge upon a single postsynaptic neuron, which allows summation of information from multiple sources. An inhibitory circuit often has inhibitory interneurons that stop the spread of a nerve signal. Reverberatory circuits have

positive feedback elements that allow a nerve impulse to continue on for a prolonged time.

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51. A) The positive increase in membrane potential to a less negative value is called the excitatory postsynaptic potential (EPSP). Because the resting membrane potential is -65 millivolts and the final membrane potential is -55 millivolts, the EPSP is $+10$ millivolts. EPSPs are always positive. Inhibitory postsynaptic potentials are always negative because the membrane potential is lowered to a more negative value.

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52. C) Activation of G proteins usually changes the long-term response characteristics of the neuron.

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53. C) Proprioceptive sensation depends on tactile and joint capsule receptors.

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54. D) Vasospasm and eventually ischemia in a sensory area of cortex is thought to be the basis for the prodromal symptoms experienced by patients with migraines.

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55. C) Both warm and cold receptors are able to respond to steady-state temperatures, as well as changes in temperature.

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56. A) The Pacinian corpuscle transmits a modality of sensation (vibration) that is transmitted in the dorsal column–medial lemniscal system. The first synaptic connection in this sensory pathway is in the dorsal column nuclei on the ipsilateral side of the body.

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57. B) Warm receptors mainly transmit signals along relatively slow-conducting type C fibers.

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58. A) Descending cortical modulation of somatosensation involves an increase or decrease in the perception of signal intensity.

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59. D) The action potential is described as an “all or none” process. Once initiated, the action potential runs its course to completion.

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60. C) A facilitated neuron is one whose resting membrane potential is closer to the threshold for activation; that is, less negative or in the depolarizing direction.

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The Nervous System: B. The Special Senses

1. A 10-year-old boy looks at ants through a magnifying glass. He finds that the ants must be 10 centimeters from the convex lens to be in focus. Which value best describes the refractive power of the lens (in diopters)?
 - A) 0.1
 - B) 1.0
 - C) 10
 - D) 100
 - E) 1000
2. Which of the following best describes the “blind spot” of the eye?
 - A) Located 5 degrees lateral to the central point of vision
 - B) The exit point of the optic nerve
 - C) Contains only rods and thus has monochromatic vision
 - D) Contains no blood vessels
 - E) The area where chromatic aberration of the lens is the greatest
3. A 6-year-old boy with albinism is taken to the ophthalmologist because of difficulty seeing. Testing shows that his visual acuity is reduced. Which of the following is the most likely cause of the decrease in visual acuity in this boy?
 - A) Cataracts
 - B) Hyperopia
 - C) Myopia
 - D) Photophobia
 - E) Presbyopia
4. A 53-year-old woman with celiac disease visits the physician because of difficulty seeing at night. The woman has frequent, foul-smelling stools. Stool analysis reveals a high content of partially digested fat. A decrease in blood levels of which of the following is the most likely cause of her night blindness?
 - A) 2-Monoglycerides
 - B) Amino acids
 - C) Free fatty acids
 - D) Glucose
 - E) Vitamin A
 - F) Vitamin B₁₂
5. Which substance will elicit the sensation of bitter taste?
 - A) Aldehydes
 - B) Alkaloids
 - C) Amino acids
 - D) Hydrogen ions
 - E) Ketones
6. Damage to the sixth cranial nerve will produce which deficit in eye movement?
 - A) Inability to move the eyes in a vertical up-and-down motion
 - B) Inability to rotate the eyes within the eye socket
 - C) Inability to move the eyes laterally toward the midline
 - D) Inability to move the eyes laterally away from the midline
 - E) Vertical strabismus
7. The condition of cataracts is usually the result of which process or condition?
 - A) Denaturation of the proteins in the lens of the eye
 - B) Elongated eye globe
 - C) Unresponsive and dilated pupil
 - D) Coagulation of the proteins in the lens of the eye
 - E) Increase in intraocular pressure
8. Which substance will elicit the sensation of sour taste?
 - A) Aldehydes
 - B) Alkaloids
 - C) Amino acids
 - D) Hydrogen ions
 - E) Ketones
9. Which taste sensation is the most sensitive (i.e., has the lowest stimulation threshold)?
 - A) Acid
 - B) Bitter
 - C) Salty
 - D) Sour
 - E) Sweet

10. An 85-year-old woman visits the ophthalmologist because of difficulty seeing. The patient is given an eye examination, and bifocal lenses are prescribed. The physician notes that the lenses of her eyes are clear. The woman sees well with her new prescription glasses. Which of the following best describes the most likely vision problem in this woman?
 - A) Cataracts
 - B) Glaucoma
 - C) Hyperopia
 - D) Myopia
 - E) Presbyopia
 11. Which of the following is the middle ear ossicle that is attached to the tympanic membrane?
 - A) Columella
 - B) Incus
 - C) Malleus
 - D) Modiolus
 - E) Stapes
 12. Light entering the eye passes through which retinal layer first?
 - A) Inner nuclear layer
 - B) Outer nuclear layer
 - C) Outer plexiform layer
 - D) Photoreceptor layer
 - E) Retinal ganglion layer
 13. A 25-year-old student with 20/20 vision looks up from his book to view his girlfriend sitting on the other side of the room. Which of the following is most likely to occur when the student changes his view from his book to his girlfriend?
 - A) Thicker lens, contraction of ciliary muscle
 - B) Thicker lens, relaxation of ciliary muscle
 - C) Thinner lens, contraction of ciliary muscle
 - D) Thinner lens, relaxation of ciliary muscle
 14. A 60-year-old woman visits the ophthalmologist because of eye pain. Tests show that her right eye has an intraocular pressure of 22 mm Hg and her left eye has an intraocular pressure of 25 mm Hg. Which of the following is the most likely cause of eye pain in this woman?
 - A) Decreased hydraulic resistance of trabecular meshwork
 - B) Decreased production of aqueous humor
 - C) Increased hydraulic resistance of trabecular meshwork
 - D) Increased production of aqueous humor
 15. Ganglion cells attached to photoreceptors located on the temporal portion of the retina project to which structure?
 - A) Contralateral lateral geniculate nucleus
 - B) Ipsilateral lateral geniculate nucleus
 - C) Ipsilateral medial geniculate nucleus
 - D) Calcarine fissure
 - E) Contralateral medial geniculate nucleus
 16. When parallel light rays pass through a concave lens, which of the following will occur?
 - A) The rays converge toward each other
 - B) The rays diverge away from each other
 - C) The rays maintain a parallel relationship
 - D) The rays reflect back in the direction from where they came
 - E) The rays refract to one focal point
 17. A 40-year-old woman is admitted emergently to the hospital because of sudden, severe pain in her right eye. Tests show an intraocular pressure of 30 mm Hg in her right eye; the intraocular pressure of her left eye is 15 mm Hg. Which of the following is the most likely cause of eye pain in this woman?
 - A) Acute angle-closure glaucoma
 - B) Chronic glaucoma
 - C) Conjunctivitis
 - D) Corneal abrasion
 - E) Open-angle glaucoma
 - F) Optic neuritis
 18. Which compartment of the cochlea contains the organ of Corti?
 - A) Ampulla
 - B) Saccule
 - C) Scala media
 - D) Scala tympani
 - E) Scala vestibuli
 19. Which molecules combine to form rhodopsin?
 - A) Bathorhodopsin and 11-cis-retinal
 - B) Bathorhodopsin and all-trans-retinal
 - C) Bathorhodopsin and scotopsin
 - D) Scotopsin and 11-cis-retinal
 - E) Scotopsin and all-trans-retinal
 20. Analysis of visual detail occurs in which secondary visual area?
 - A) Brodmann's area 18
 - B) Inferior ventral and medial regions of the occipital and temporal cortex
 - C) Frontal lobe
 - D) Occipitoparietal cortex
 - E) Posterior midtemporal area
- Questions 21–23**
- A 23-year-old student is trapped in an elevator with no light. Twenty minutes later the student finds an emergency light and turns it on. Use this information to answer Questions 21–23.
21. Which substance is most likely to increase in the rods of the retina when the light is turned on?
 - A) Cyclic adenosine monophosphate (cAMP)
 - B) Cyclic guanosine monophosphate (cGMP)
 - C) Metarhodopsin II
 - D) Rhodopsin
 - E) Vitamin A

22. Which of the following best describes the permeability to sodium and potassium in rod cells in response to the light?
- Decreased sodium permeability, decreased potassium permeability
 - Decreased sodium permeability, increased potassium permeability
 - Decreased sodium permeability, no change in potassium permeability
 - Increased sodium permeability, decreased potassium permeability
 - Increased sodium permeability, increased potassium permeability
 - Increased sodium permeability, no change in potassium permeability
23. Which of the following best describes the electrical response of the rods to light?
- Action potential
 - Capacitive discharge
 - Depolarization
 - Hyperpolarization
24. Which substance is responsible for the umami taste sensation?
- Acetic acid
 - Potassium tartrate
 - Long-chained organic substances containing nitrogen
 - Fructose
 - Glutamate
25. Which cell type(s) have action potentials in the retina of the human eye?
- Bipolar cells and ganglion cells
 - Bipolar cells only
 - Bipolar cells, horizontal cells, and ganglion cells
 - Ganglion cells and horizontal cells
 - Ganglion cells only
 - Horizontal cells only
26. Olfactory receptor cells belong to which group of cells?
- Bipolar neurons
 - Fibroblasts
 - Modified epithelial cells
 - Multipolar neurons
 - Pseudounipolar neurons
27. Which of the following best describes when the transmission of sound waves in the cochlea occurs?
- When the foot of the stapes moves inward against the oval window and the round window bulges outward
 - When the foot of the stapes moves inward against the round window and the oval window bulges outward
 - When the head of the malleus moves inward against the oval window and the round window bulges outward
 - When the incus moves inward against the oval window and the round window bulges outward
 - When the incus moves inward against the round window and the oval window bulges outward
28. Under low or reduced light conditions, which chemical compound is responsible for the inward-directed sodium current in the outer segments of the photoreceptors?
- Metarhodopsin II
 - cGMP
 - 11-cis retinal
 - cAMP
 - 11-trans retinal
29. Which cells in layer IV of the primary visual cortex detect orientation of lines and borders?
- Border cells
 - Complex cells
 - Ganglion cells
 - Hypercomplex cells
 - Simple cells
- Questions 30 and 31**
A 20-year-old soldier sustains a noise-induced hearing loss over a period of 6 months from multiple exposures to loud sounds. Use this information to answer Questions 30 and 31.
30. Loss of which structure is most likely to contribute to the hearing deficit?
- Cochlea
 - Inner hair cells
 - Organ of Corti
 - Scala media
 - Scala vestibuli
31. An increase in which of the following is the most likely cause of this hearing loss?
- Connexin 26
 - Endolymph
 - Perilymph
 - Reactive oxygen species
32. Which event occurs in photoreceptors during photo-transduction in response to light?
- Phosphodiesterase activity decreases
 - Transducin activity decreases
 - Hydrolysis of cGMP increases
 - Neurotransmitter release increases
 - The number of open voltage-gated calcium channels increases
33. During photoreception, all the following increase except
- cGMP phosphodiesterase
 - Transducin
 - cAMP
 - Metarhodopsin II
 - Sodium influx into the outer segment of the rod

Questions 34 and 35

A 50-year-old woman visits an otolaryngologist for sudden bouts of dizziness that subside after about 20 minutes. She also has temporary hearing losses and a feeling of fullness in her right ear; low-pitched buzzing sounds occur intermittently in her right ear. Physical examination shows nystagmus during a dizzy spell. Use this information to answer Questions 34 and 35.

34. Which of the following is the most likely diagnosis?

- A) Acoustic neuroma
- B) Aural polyp
- C) Exostosis
- D) Incus erosion
- E) Meniere's disease

35. An increase in which of the following is the most likely cause of this patient's condition?

- A) Endolymph pressure only
- B) Endolymph volume only
- C) Endolymph volume and pressure
- D) Perilymph pressure only
- E) Perilymph volume only
- F) Perilymph volume and pressure

36. The condition of myopia is usually corrected by which type of lens?

- A) Compound lens
- B) Convex lens
- C) Spherical lens
- D) Concave lens
- E) Cylindrical lens

37. Which lobe of the cerebral cortex contains the small bilateral cortical area that controls voluntary fixation movements?

- A) Frontal
- B) Limbic
- C) Occipital
- D) Parietal
- E) Temporal

38. Which sensory system has the smallest range of intensity discrimination?

- A) Auditory
- B) Gustatory
- C) Olfactory
- D) Somatosensory
- E) Visual

39. Which molecules move from the endolymph into the stereocilia and depolarize the hair cell?

- A) Calcium ions
- B) Chloride ions
- C) Hydrogen ions
- D) Potassium ions
- E) Sodium ions

40. The stereocilia of hair cells are embedded in which membrane?

- A) Basilar
- B) Reissner's
- C) Tectorial
- D) Tympanic
- E) Vestibular

41. Which cranial nerve is correctly paired with the extraocular muscle it innervates?

- A) Abducens nerve—medial rectus
- B) Oculomotor nerve—inferior oblique
- C) Oculomotor nerve—lateral rectus
- D) Oculomotor nerve—superior oblique
- E) Trochlear nerve—superior rectus

42. After olfactory receptor cells bind odor molecules, a sequence of intracellular events occurs that culminates in the entrance of specific ions that depolarize the olfactory receptor cell. Which ions are involved?

- A) Calcium
- B) Chloride
- C) Hydrogen
- D) Potassium
- E) Sodium

43. For the eye to adapt to intense light, which of the following may occur?

- A) Bipolar cells will continuously transmit signals at the maximum rate possible
- B) Photochemicals in both rods and cones will be reduced to retinal and opsins
- C) The levels of rhodopsin will be very high
- D) The size of the pupil will increase
- E) Vitamin A will convert into retinal

44. In the central auditory pathway, which option represents the correct sequence of structures in the pathway?

- A) Cochlear nuclei—superior olive—inferior colliculus via the lateral lemniscus—medial geniculate—auditory cortex
- B) Cochlear nuclei—inferior olive—inferior colliculus via the medial lemniscus—medial geniculate—auditory cortex
- C) Cochlear nuclei—superior olive—superior colliculus via the lateral lemniscus—lateral geniculate—auditory cortex
- D) Cochlear nuclei—inferior olive—inferior colliculus via the lateral lemniscus—lateral geniculate—auditory cortex
- E) Cochlear nuclei—trapezoid body—dorsal acoustic stria—inferior colliculus via the lateral lemniscus—medial geniculate—auditory cortex

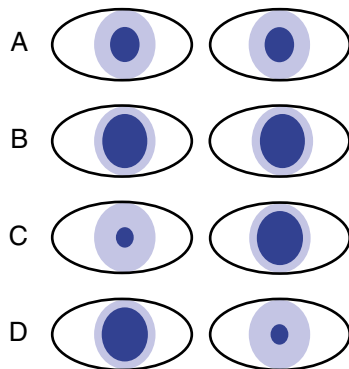
45. Which statement regarding the transmission of auditory information from the ear to the cerebral cortex is correct?
- Inferior colliculus neurons synapse in the cochlear nuclei of the brain stem
 - Neurons with cell bodies in the spiral ganglion of Corti synapse in the inferior colliculus
 - The majority of neurons from the cochlear nuclei synapse in the contralateral superior olivary nucleus
 - There is no crossing over of information between the right and left auditory pathways in the brain stem
 - Trapezoid neurons synapse in the cochlear nuclei of the brain stem
46. Which statement regarding color vision is correct?
- Green is perceived when only green cones are stimulated
 - The stimulation ratio of the three types of cones allows specific color perception
 - The wavelength of light corresponding to white is shorter than that corresponding to blue
 - When no stimulation of red, green, or blue cones occurs, there will be the sensation of seeing white
 - Yellow is perceived when green and blue cones are stimulated equally
47. Which event prompts the auditory system to interpret a sound as loud?
- A decreased number of inner hair cells become stimulated
 - A decreased number of outer hair cells become stimulated
 - Hair cells excite nerve endings at a diminished rate
 - The amplitude of vibration of the basilar membrane decreases
 - The amplitude of vibration of the basilar membrane increases
48. Which statement is correct concerning the elements of the retina?
- The total number of cones in the retina is much greater than the total number of rods
 - Each individual cone responds to all wavelengths of light
 - Photoreceptor activation (rods and cones) results in hyperpolarization of the receptor
 - The central fovea contains only rods
 - The pigment layer of the retina contains the photoreceptors
49. The condition of hyperopia is usually caused by which anomaly of the eye?
- Decreased production of melanin
 - Uneven curvature of the cornea
 - An eyeball that is shorter than normal
 - An eyeball that is longer than normal
 - A lens system that is too powerful and focuses the object in front of the retina
50. Which statement regarding the two types of deafness is correct?
- An audiogram of a person with conduction deafness would show much greater loss for air conduction than for bone conduction of sound
 - An audiogram of a person with nerve deafness would show much greater loss for bone conduction than for air conduction of sound
 - Conduction deafness occurs when the cochlea or cochlear nerve is impaired
 - Nerve deafness occurs when the physical structures that conduct the sound into the cochlea are impaired
 - Prolonged exposure to very loud sounds is more likely to cause deafness for high-frequency sounds than for low-frequency sounds
51. When a person turns the head to the left about the axis of the neck, the motion begins when the chin is directly over the right shoulder and ends with the chin directly over the left shoulder. Which option best describes the eye movements associated with this type of head rotation in a normal person?
- While the head is turning, the eyes will be moving to the right and saccadic eye motion will be to the left
 - While the head is turning, the eyes will be moving in the same direction as the head rotation and the saccadic eye motion will be to the left
 - While the head is turning, the eyes will be moving to the right and the saccadic eye motion will be to the right
 - While the head is turning, the eyes will remain stationary within the orbits and the saccadic eye motion will be to the right
 - While the head is turning, the eyes will be moving to the left and the saccadic eye motion will be to the right
52. Olfactory information transmitted to the orbitofrontal cortex passes through which thalamic nucleus?
- Dorsomedial
 - Lateral geniculate
 - Medial geniculate
 - Ventral posterolateral
 - Ventral posteromedial
53. A 29-year-old student with 20/20 vision looks at a beautiful scene. The axons of ganglion cells transmitting visual signals in the form of action potentials to the primary visual cortex are most likely to synapse in which structure?
- Lateral geniculate nucleus
 - Medial geniculate nucleus
 - Optic chiasm
 - Optic radiation
 - Superior cervical ganglion
 - Superior colliculus

54. The function of the round window can best be described by which statement?
- A) It provides the connection point for the stapes
 - B) It serves to damp out low frequency sounds such as your own voice
 - C) It transmits the frequency information into the cochlea from the tympanic membrane
 - D) It serves as the pressure relief valve for the cochlea**
 - E) It transmits amplitude information into the cochlea from the tympanic membrane
55. Which muscle is contracted as part of the pupillary light reflex?
- A) Ciliary muscle
 - B) Pupillary dilator muscle
 - C) Pupillary sphincter muscle**
 - D) Radial fibers of the iris
 - E) Superior oblique muscle

Questions 56 and 57

A 24-year-old woman sustains a laceration on the right side of the neck in a motor vehicle accident. Physical examination shows that her right pupil is constricted, her right eyelid droops, the skin is dry on the right side of her face, and the conjunctiva of her right eye is red. Use this information to answer Questions 56 and 57.

56. What is the most likely diagnosis?
- A) Cone-rod dystrophy
 - B) Horner's syndrome
 - C) Iris heterochromia
 - D) Retinoblastoma
 - E) Xerophthalmia



57. Which test result shown in the above figure is most likely after topical treatment with cocaine in both eyes?
- A) A
 - B) B
 - C) C
 - D) D

58. Which neurotransmitter is released by both rods and cones at their synapses with bipolar cells?
- A) Acetylcholine
 - B) Dopamine
 - C) Glutamate**
 - D) Glycine
 - E) Serotonin
59. Which of the following allows the visual apparatus to accurately determine the distance of an object from the eye (depth perception)?
- A) Monocular vision
 - B) The location of the retinal image on the retina
 - C) The phenomenon of stationary parallax
 - D) The phenomenon of stereopsis**
 - E) The size of the retinal image if the object is of unknown size
60. Which of the following provides about two thirds of the 59 diopters of refractive power of the eye?
- A) Anterior surface of the cornea**
 - B) Anterior surface of the lens
 - C) Iris
 - D) Posterior surface of the cornea
 - E) Posterior surface of the lens
61. Which photoreceptor responds to the broadest spectrum of wavelengths of light?
- A) Rod receptors
 - B) Green cone receptors
 - C) Blue cone receptors
 - D) Red cone receptors
 - E) Cells containing melanin in the pigment layer
62. Which structure secretes the intraocular fluid of the eye?
- A) Ciliary processes**
 - B) Cornea
 - C) Iris
 - D) Lens
 - E) Trabeculae
63. Which type of papillae is located in the posterior part of the tongue?
- A) Circumvallate**
 - B) Foliate
 - C) Fungiform
 - D) Fungiform and circumvallate
 - E) Papilla of Vater

64. Which statement regarding retinal ganglion cells is correct?
- One W ganglion cell from the periphery of the retina typically transmits information from one rod
 - One X ganglion cell from the fovea typically transmits information from as many as 200 cones
 - W ganglion cells respond best to directional movement or vision under very bright conditions
 - X ganglion cells respond best to color images and are the most numerous of the three types of ganglion cells
 - Y ganglion cells respond best to rapid changes in the visual image and are the most numerous of the three types of ganglion cells
65. Auditory information is relayed through which thalamic nucleus?
- Dorsomedial
 - Lateral geniculate
 - Medial geniculate
 - Ventral posterolateral
 - Ventral posteromedial
66. Which of the following describes the phenomenon of taste preference?
- A central nervous system process
 - The result of neonatal stimulation of circumvallate papilla
 - A learned behavior in animals
 - A result of taste bud maturation
 - A result of taste bud proliferation after exposure to glutamic acid
67. The primary auditory cortex lies primarily in which lobe of the cerebral cortex?
- Frontal
 - Limbic
 - Occipital
 - Parietal
 - Temporal
68. The first central synapse for neurons transmitting the sweet taste sensation is in which structure?
- Dorsal sensory nucleus of vagus nerve
 - Nucleus of solitary tract
 - Nucleus of olfactory nerve
 - Nucleus of hypoglossal nerve
 - Nucleus of facial nerve
69. Which statement best describes the underlying basis of the dark current in the outer segment of the photoreceptors?
- Dark current results from the influx of sodium ions via c-AMP-dependent sodium channels
 - Dark current results from the influx of sodium ions via c-GMP-dependent sodium channels
 - Dark current results from the efflux of potassium ions via c-GMP-dependent potassium channels
 - Dark current results from the efflux of sodium ions via c-GMP-dependent sodium channels
 - Dark current results from the efflux of sodium ions via c-AMP-dependent sodium channels
70. Which structure functions to ensure that each of the three sets of extraocular muscles is reciprocally innervated so that one muscle of the pair relaxes while the other contracts?
- Edinger-Westphal nucleus
 - Medial longitudinal fasciculus
 - Pretectal nucleus
 - Superior colliculus
 - Suprachiasmatic nucleus
71. The intraocular fluid of the eye flows from the canal of Schlemm into which location?
- Anterior chamber
 - Aqueous veins
 - Lens
 - Posterior chamber
 - Trabeculae
72. Which retinal cells have action potentials?
- Amacrine cells
 - Bipolar cells
 - Ganglion cells
 - Horizontal cells
 - Photoreceptors
73. Which brain stem structure plays a major role in determining the direction from which a sound originates?
- Cochlear nucleus
 - Inferior colliculus
 - Lateral lemniscus
 - Superior olivary nucleus
 - Trapezoid
74. A 25-year-old student studies for a test in medical physiology. The visual contrast of the subject matter is enhanced due to lateral inhibition of the visual input by which cell type in the retina?
- Amacrine cells
 - Bipolar cells
 - Ganglion cells
 - Horizontal cells
75. Which type of papillae is located in the folds along the lateral surfaces of the tongue?
- Circumvallate
 - Foliate
 - Fungiform
 - Fungiform and circumvallate
 - Papilla of Vater

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1. **C)** The refractive power of a lens (in diopters) = 1 meter/focal length; if the subject matter is in focus when a convex lens is 1 meter from the subject matter, the lens has a refractive power of 1 meter/1 meter = 1 diopter. Thus, there is an inverse relationship between focal length and refractive power; a thicker convex lens has a shorter focal length and a greater refractive power. In this problem, the lens must be 10 centimeters from the subject matter to be in focus (focal length = 100 millimeters); therefore, 1000 millimeters/100 millimeter = 10 diopters. Because the retina of the eye is about 17 millimeters behind the lens, the refractive power of the lens of the eye is about 59 diopters.
TMP13 pp. 637-638
2. **B)** The blind spot is located 15 degrees lateral to the central point of vision. It is the location where fibers that make up the optic nerve exit the globe of the eye. There are no photoreceptors in this location.
TMP13 p. 665
3. **D)** Photophobia is discomfort or pain to the eyes due to light exposure; it is a medical condition, not a fear or phobia. The lack of melanin (black pigment) in the irises of the eyes makes them somewhat translucent, so they cannot block light effectively. The lack of melanin in the pigment layer of the retina causes light to scatter inside the globe of the eye, which decreases contrast and visual acuity.
TMP13 pp. 648-649
4. **E)** Vitamin A is a fat-soluble vitamin that can be excreted in the feces along with fat in persons with celiac disease and other diseases that cause malabsorption of intestinal contents. A lack of vitamin A can cause a decrease in production of retinal, which is necessary for synthesis of rhodopsin in the rods of the retina. Decreased levels of rhodopsin in the rods can lower the sensitivity of the retina to light, thus causing night blindness.
TMP13 pp. 648-649
5. **B)** The taste sensation of bitter is caused by many organic substances that contain nitrogen, as well as by alkaloids.
TMP13 p. 685
6. **D)** The sixth cranial nerve is also known as the *abducens nerve*. The abducens nerve innervates the lateral rectus muscle, which is attached to the lateral surface of the globe of the eye. Contraction of the lateral rectus muscle results in movement of the eyeball laterally away from the midline of the face in an abducting manner—thus the name *abducens nerve*.
TMP13 pp. 665-666
7. **D)** The condition of cataracts causes the lens of the eye to become opaque and resemble the look of water in a waterfall or rapids in a river, thus the name, cataract. A cataract results from the progressive coagulation of the proteins that make up the lens. One can think of this coagulation as similar to the white of an egg turning opaque as it is cooked. Heating the egg white results in coagulation of the proteins contained within it.
TMP13 p. 642
8. **D)** The taste sensation of sour is proportional to the logarithm of the hydrogen ion concentration caused by acids. The taste sensation of sweet is caused by a long list of chemicals, including sugars, alcohols, aldehydes, ketones, and amino acids.
TMP13 p. 685
9. **B)** The bitter taste sense is much more sensitive than the other sensations because it provides an important protective function against many dangerous toxins in food.
TMP13 p. 686
10. **E)** A person with presbyopia cannot accommodate for near and far vision, which means that the lenses of the eyes have lost their elasticity and thus cannot change their focal point. A child with good vision has 14 diopters of accommodation; this accommodation decreases throughout life until, at approximately 70 years, the lenses cannot change their shape and the power of accommodation is then zero. People with zero power of accommodation are said to be presbyopic.
TMP13 p. 640
11. **C)** The malleus is attached to the tympanic membrane, and the stapes is attached to the oval window. The incus has articulations with both of these bones.
TMP13 p. 673
12. **E)** Light passes through the eye to the retina in the posterior portion of the eye. The most anterior layer of the retina, through which light passes first, is the retinal ganglion layer. Light then passes through the other cell layers of the retina until it reaches the photoreceptors in the posterior region of the retina.
TMP13 p. 647
13. **D)** Light rays from distant objects do not require as much refraction (bending) as do light rays from objects close at hand. Therefore, a thinner lens with less curvature is required for viewing distant objects. The process of accommodation adjusts the thickness of the lens for near and far vision by contracting or relaxing the ciliary muscle that surrounds the lens of the eye; contraction

of the ciliary muscle thickens the lens, and relaxation causes the lens to become thinner.

TMP13 pp. 639-640

- 14. C)** This woman has open-angle glaucoma, which is the most common type of glaucoma. Glaucoma is the second leading cause of blindness worldwide after cataracts. Blindness occurs because of damage to the optic nerve. The high intraocular pressure causes blood vessels and axons of the optic nerve to be compressed at the optic disc, which leads to poor nutrition with possible death of the neurons. The main cause of open-angle glaucoma is reduced flow of aqueous humor through the trabecular meshwork because of tissue debris, white blood cells, deposition of fibrous material, and other factors that increase the hydraulic resistance of the meshwork.

TMP13 p. 646

- 15. B)** The axons of the ganglion cells make up the fibers of the optic nerve. The first synapse in the visual system takes place in the lateral geniculate nucleus. Ganglion cells attached to photoreceptors on the temporal side of the retina project to the same-sided or ipsilateral lateral geniculate nucleus. Fibers from the nasal side of the retina cross over to the opposite or contralateral lateral geniculate nucleus in the optic chiasm. The medial geniculate nucleus is a sensory relay for the auditory system.

TMP13 p. 661

- 16. B)** A concave lens diverges light rays; in contrast, a convex lens will converge light rays toward each other. If a convex lens has the appropriate curvature, parallel light rays will be bent so that all pass through a single point, called the *focal point*.

TMP13 p. 636

- 17. A)** This woman has acute angle-closure glaucoma, which is a medical emergency. Sudden closure of the iridocorneal angle prevents aqueous humor from reaching its outflow pathway in the canal of Schlemm. Intraocular pressure can increase rapidly and cause blindness without immediate treatment. Chronic glaucoma and open-angle glaucoma are the same disease. Conjunctivitis (pinkeye) is an inflammation of the conjunctiva. A corneal abrasion is a scratch on the cornea; it can be very painful. Optic neuritis is inflammation of the optic nerve.

TMP13 p. 646

- 18. C)** The ampulla and saccule are part of the vestibular apparatus, not the cochlear apparatus. The cochlea has three main compartments, with fluid movement occurring in the scala vestibuli and scala media in response to sound vibrations. The organ of Corti is contained within the scala media.

TMP13 pp. 674-675

- 19. D)** Rhodopsin is the light-sensitive chemical in rods. Scotopsin and all-trans retinal are the breakdown products of rhodopsin, which has absorbed light energy. The all-trans retinal is converted into 11-cis retinal, which can recombine with scotopsin to form rhodopsin.

TMP13 pp. 649-650

- 20. B)** Visual information from the primary visual cortex (Brodman's area 17) is relayed to Brodman's area 18 and then into other areas of the cerebral cortex for further processing. Analysis of three-dimensional position, gross form, and motion of objects occurs in the posterior midtemporal area and occipitoparietal cortex. Analysis of visual detail and color occurs in the inferior ventral and medial regions of the occipital and temporal cortex.

TMP13 pp. 662-664

- 21. C)** Photons activate rhodopsin to become metarhodopsin II in the rods of the retina, which means that exposure to light decreases the concentration of rhodopsin and increases the concentration of metarhodopsin II, also called *activated rhodopsin*. Metarhodopsin II leads to a decrease in cGMP through a series of biochemical reactions; cAMP levels are unchanged. Vitamin A levels are not likely to change with exposure to light.

TMP13 pp. 647, 649-651

- 22. C)** Activated rhodopsin (metarhodopsin II) closes cGMP-gated sodium channels by lowering levels of cGMP; this action decreases sodium permeability. Potassium permeability is not affected.

TMP13 pp. 650-651

- 23. D)** Exposure of rods to light causes cGMP-gated sodium channels in the cell membrane to close; this action causes rods to hyperpolarize from a resting value of about -40 millivolts to as low as -70 millivolts. Action potentials do not occur in rods or cones. A capacitive discharge occurs during the course of an action potential.

TMP13 p. 651

- 24. E)** The term *umami* is derived from the Japanese word for savory or delicious and is often described as similar to the taste of meat. Glutamate is the chemical believed to elicit the umami taste sensation.

TMP13 p. 686

- 25. E)** Ganglion cells are the only cell type in the retina that have action potentials. The axons of ganglion cells comprise the optic nerve. Bipolar cells, cones, rods, horizontal cells, and other cell types in the retina signal information by electrotonic conduction, which allows a graded response proportional to light intensity.

TMP13 pp. 655, 656, 658

- 26. A)** The receptor cells for the smell sensation are bipolar nerve cells derived originally from the central nervous system itself.

TMP13 p. 689

- 27. A)** The malleus is connected to the tympanic membrane, the incus articulates with the malleus and stapes, and the stapes is connected to the oval window.
TMP13 p. 673
- 28. B)** In low light conditions, the level of cGMP is high. cGMP-dependent sodium channels in the outer portions of the rods and cones allow sodium ions to pass from the extracellular space to the intracellular space of the photoreceptor. This passage results in a membrane potential that is somewhat lower than the resting membrane potential of a typical neuron. The movement of the sodium ions and resulting electrical potential change as a result of this enhanced permeability is known as the *dark current*.
TMP13 pp. 650-651
- 29. E)** The simple cells of the primary visual cortex detect orientation of lines and borders, whereas the complex cells detect lines oriented in the same direction but are not position specific. That is, the line can be displaced moderate distances laterally or vertically, and the same few neurons will be stimulated as long as the line is the same direction.
TMP13 p. 664
- 30. B)** Noise-induced hearing loss (NIHL) is the most common acquired cause of hearing loss worldwide. NIHL is usually caused by damage and eventual death of the inner hair cells located in the organ of Corti of the cochlea; these cells do not grow back. The inner hair cells are the actual sensory receptors of the organ of Corti. The scala media and scala vestibuli are fluid-filled coiled tubes that comprise the cochlea.
TMP13 pp. 682, 683
- 31. D)** Prolonged exposure to excessive sound levels or loud sounds overstimulates hair cells, causing them to produce large amounts of reactive oxygen species, which can cause oxidative cell death. Animal studies have shown that antioxidant vitamins administered the day after noise exposure can reduce the hearing loss, but pretreatment is more effective. Low levels of connexin 26 due to gene mutation are thought to constitute a congenital hearing loss. Perilymph is the fluid contained in the scala vestibuli and scala tympani of the cochlea; endolymph is the fluid contained in the scala media and membranous labyrinth.
TMP13 pp. 682-683
- 32. C)** In the dark state, cGMP helps maintain the open state of the sodium channels in the outer membrane of the rod. Hydrolysis of cGMP by light causes these sodium channels to close. Less sodium is able to enter the rod outer segment, thus hyperpolarizing the rod.
TMP13 pp. 651-652
- 33. E)** During photoreception, the active compound metarhodopsin is formed, which in turn activates a G protein called *transducin*. The transducin activates a cGMP phosphodiesterase that destroys cGMP. cGMP-dependent sodium channels close, and the influx of sodium ions into the outer segment of the photoreceptors decreases.
TMP13 pp. 650-652
- 34. E)** This woman has Meniere's disease, which is a disorder of the inner ear that affects hearing and balance. The disease results from excess endolymph in the scala media and membranous labyrinth. The cause is not known, but it appears to have a genetic component. Symptoms include vertigo, nystagmus, low-pitched tinnitus, and sudden but temporary hearing loss; hearing loss can become permanent. Acoustic neuroma is a slow-growing benign tumor that develops on the auditory nerve. An aural polyp is a growth in the auditory canal that may be attached to the tympanic membrane, or it may grow from the middle ear. An exostosis is the formation of new bone on the surface of an existing bone; it sometimes occurs in the auditory canal of swimmers after prolonged exposure to cold water and is sometimes called "surfer's ear." The incus bone is anvil-shaped and is one of the three ossicles in the middle ear.
TMP13 pp. 677, 678
- 35. C)** Increases in both volume and pressure of endolymph in the membranous labyrinth produce the symptoms of Meniere's disease; the reason for this buildup of endolymph is unknown. The membranous labyrinth is composed mainly of the cochlea and balance organs (semicircular canals, utricle, and saccule). Repeated rupturing and healing of the endolymphatic sac of the membranous labyrinth can account for the intermittent symptoms of Meniere's disease. The endolymphatic sac is thought to regulate hydrostatic pressure of endolymph by simple expansion or collapse; it may also have secretory and absorption functions.
TMP13 pp. 677, 678
- 36. D)** In myopia the focal point of the lens system of the eye is in front of the retina. A concave lens will diverge light rays. By placing the proper concave lens in front of the eye, the divergence of light rays will move the focal point from in front of the retina to a position on the retina.
TMP13 p. 641
- 37. A)** A bilateral premotor cortical region of the frontal lobes controls voluntary fixation movements. A lesion of this region makes it difficult for a person to "unlock" their eyes from one point of fixation and then move them to another point.
TMP13 pp. 666-667

38. C) Concentrations that are only 10 to 50 times above threshold values will evoke maximum intensity of smell, which is in contrast to most other sensory systems of the body, where the range of intensity discrimination may reach 1 trillion to 1. This phenomenon can perhaps be explained by the fact that smell is concerned more with detecting the presence or absence of odors than with quantitative detection of their intensities.

TMP13 p. 690

39. D) Although most cells in the nervous system depolarize in response to sodium entry, hair cells are one group of cells that depolarize in response to potassium entry.

TMP13 p. 677

40. C) The scala media is bordered by the basilar membrane and Reissner's membrane and contains a tectorial membrane. The apical border of hair cells has stereocilia that are embedded in the tectorial membrane.

TMP13 p. 677

41. B) The abducens nerve innervates the lateral rectus muscle. The trochlear nerve innervates the superior oblique muscle. The oculomotor nerve innervates the medial rectus, inferior oblique, superior rectus, and inferior rectus muscles.

TMP13 p. 666

42. E) Even the minutest concentration of a specific odorant initiates a cascading effect that opens extremely large numbers of sodium channels. This phenomenon accounts for the exquisite sensitivity of the olfactory neurons to even the slightest amount of odorant.

TMP13 p. 690

43. B) The reduction of rhodopsin and cone pigments by light reduces the concentrations of photosensitive chemicals in rods and cones. Thus, the sensitivity of the eye to light is correspondingly reduced. This phenomenon is called *light adaptation*.

TMP13 pp. 652-653

44. A) Auditory fibers enter the cochlear nucleus. Fibers from the cochlear nucleus pass to the inferior colliculus via the lateral lemniscus. Fibers from the inferior colliculus travel to the medial geniculate nucleus and from there to the primary auditory cortex.

TMP13 p. 679

45. C) Neurons with cell bodies in the spiral ganglion of Corti synapse in the cochlear nuclei. The majority of the cochlear nuclei neurons synapse in the contralateral superior olivary nucleus. Crossing over occurs in at least three places in the pathway, and a preponderance of auditory transmission is in the contralateral pathway. From the superior olivary nucleus, the auditory pathway then passes upward through the lateral lemniscus, with most auditory fibers terminating at the inferior colliculus. From there, the pathway continues on to the medial geniculate nucleus and then to the primary auditory cortex.

TMP13 p. 679

46. B) Research has shown that the nervous system perceives the sensation of a specific color by interpreting the set of ratios of stimulation of the three types of cones. Investigators used only red, green, and blue monochromatic lights mixed in different combinations. All gradations of colors the human eye can detect were detected with only these three colors.

TMP13 pp. 653-654

47. E) The auditory system determines loudness in at least three ways. First, the amplitude of vibration of the basilar membrane increases so that hair cells excite nerve endings at more rapid rates. Second, more and more hair cells on the fringes of the resonating portion of the basilar membrane become stimulated. Third, outer hair cells become recruited at a significant rate.

TMP13 pp. 676, 678

48. C) Unlike most other sensory receptors that depolarize when activated, the photoreceptors produce the opposite response, which is hyperpolarization. The total number of rods is much greater than the number of cones. Cones respond to a very specific range of wavelengths of light. The pigment layer is posterior to the retinal layer that contains the photoreceptors.

TMP13 p. 650

49. C) In hyperopia the focal point of the eye's lens system is behind the retina. This is usually the result of an eyeball that is too short in the anterior to posterior direction.

TMP13 p. 640

50. A) With nerve deafness, there is damage to the cochlea, auditory nerve, or neural pathway. The ability to hear sound as tested by both air conduction and bone conduction is greatly reduced or lost with nerve deafness. However, with conduction deafness, the person retains the ability to hear sound by bone conduction, but not by air conduction.

TMP13 p. 682

51. A) In the situation described, the eyes will fix on an object in the visual field and remain on that object while the head is turning to the left, resulting in eye movement to the right as the head is turned to the left. When the object is no longer in the central field of vision, the eyes will exhibit a quick jumping movement to the left (i.e., in the direction of the head rotation) and fix on a new object in the visual field. This jump is called a *saccade*. This process will repeat until the head has turned all the way to the left. During saccadic eye movement, vision is suppressed.

TMP13 pp. 667-668

52. A) A newer olfactory pathway has been found that projects to the dorsomedial thalamic nucleus and then to the orbitofrontal cortex. However, the older olfactory pathways bypass the thalamus to reach the cortex, in contrast to other sensory systems, which have thalamic relays.

TMP13 p. 691

- 53. A)** Ganglion cells of the retina have synaptic connections within the lateral geniculate nucleus (LGN); from there the visual signals (action potentials) are transmitted to the primary visual cortex. Ganglion cells in the nasal half of the retina synapse in the contralateral LGN, whereas ganglion cells from the temporal half of the retina synapse in the ipsilateral LGN. Decussation occurs in the optic chiasm. Postsynaptic neurons in the LGN travel in the optic radiations and synapse in a fan-shaped manner in the primary visual cortex.

TMP13 pp. 661-662

- 54. D)** The cochlea is a structure of tubes and chambers that is filled with fluid. The fluid is not compressible. As the stapes moves back and forth against the oval window, the increase and decrease in pressure caused by that in-and-out movement of the oval window is relieved by the opposite back-and-forth movement of the round window.

TMP13 p. 675

- 55. C)** In a normal individual, shining a light in either eye will result in both pupils constricting due to contraction of the pupillary sphincter muscles. In contrast, the pupillary dilator muscle dilates the pupil. The ciliary muscle is involved in focusing the eye (accommodation).

TMP13 p. 672

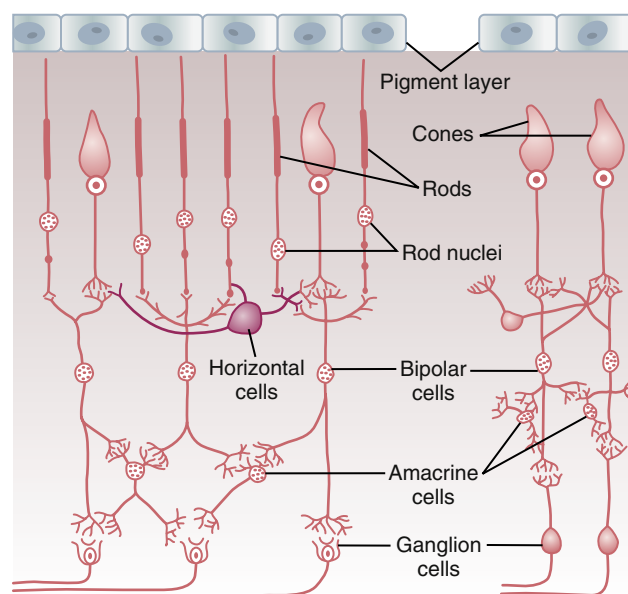
- 56. B)** This woman has Horner's syndrome, which is not a disease but rather a symptom of a disease or other problem. In this problem, lacerations to the right side of the neck have damaged the sympathetic nerves to the right eye and right side of the face. Other causes of Horner's syndrome include aortic dissection that compresses adjacent tissues, carotid dissection, Pancoast lung tumor, and tuberculosis, and it can also be congenital. Disruption of sympathetic nerves to the eye causes ipsilateral miosis, ptosis, and dilated blood vessels in the conjunctiva. Cone-rod dystrophy is a chronic disease in which the rods and cones deteriorate over time. Iris heterochromia is a difference in the color of the irises of the two eyes, which often occurs in persons with Horner's syndrome before the age of 2 years but not in adults, in whom eye color has been established. Retinoblastoma is a cancer of the eye in children. Xerophthalmia (also called *dry eye syndrome*) is a disease caused by dryness of the eye.

TMP13 p. 670

- 57. D)** Cocaine blocks the reuptake of norepinephrine, increasing its concentration at the nerve terminal. Norepinephrine relaxes the pupillary dilator muscle (also called the *pupillary sphincter*), causing the pupil to become larger. Failure of cocaine to cause pupillary dilation indicates disruption of the sympathetic nerves to the pupillary dilator muscle because norepinephrine is not being released at the nerve-muscle junction.

A more recent approach is to apply an α -adrenergic agonist (such as apraclonidine) to both eyes. The pupillary dilator muscle responds to denervation by increasing the number of its α -1 receptors. The weak α -1 adrenergic properties of apraclonidine have no effect on the normal pupillary dilator muscle but cause extensive dilation of the hypersensitive, denervated pupillary dilator muscle. Thus, with application of apraclonidine, the correct answer would be C because the right eye is denervated and thus hypersensitive to α -1 adrenergic stimulation.

TMP13 p. 670



- 58. C)** At least eight types of neurotransmitter substances have been identified for amacrine cells. The neurotransmitters used for bipolar and horizontal cells are unclear, but it is well established that rods and cones release glutamate at their synapses with bipolar cells (see figure above).

TMP13 p. 655

- 59. D)** Because one eye is a little more than 2 inches to the side of the other eye, the images on the two retinas differ from one another. This binocular parallax (stereopsis) allows a person with two eyes far greater ability than a person with only one eye to judge relative distances when objects are nearby.

TMP13 p. 644

- 60. A)** The principal reason why the anterior surface of the cornea provides most of the refractive power of the eye is that the refractive index of the cornea is markedly different from that of air.

TMP13 p. 638

- 61. D)** Intuitively, one would guess that the rod photoreceptor would have the greatest range of spectral sensitivity. However, it is the red cone that has the broadest spectral sensitivity, followed by the rods, the green cones, and finally the blue cones, which have the narrowest range of spectral sensitivity.

TMP13 p. 652

- 62. A)** Ciliary processes secrete all the aqueous humor of the intraocular fluid at an average rate of 2 to 3 $\mu\text{l}/\text{min}$. These processes are linear folds that project from the ciliary muscle into the space behind the iris. The intraocular fluid flows from behind the iris through the pupil into the anterior chamber of the eye.

TMP13 p. 645

- 63. A)** Circumvallate papillae are located in the posterior part of the tongue, fungiform papillae in the anterior part of the tongue, and foliate papillae on the lateral part of the tongue. The papilla of Vater empties pancreatic secretions and bile into the duodenum.

TMP13 p. 687

- 64. D)** There are three distinct groups of retinal ganglion cells, designated as W, X, and Y cells. W cells transmit rod visual signals. Y cells are the least numerous and transmit information about rapid changes in the visual image. X cells are the most numerous and receive input from cones regarding the visual image and color vision.

TMP13 pp. 657-658

- 65. C)** The medial geniculate nucleus is the thalamic nucleus that conveys auditory information from the brain stem to the primary auditory cortex.

TMP13 p. 679

- 66. A)** Taste preference, although not completely understood, is believed to involve a central process.

TMP13 p. 688

- 67. E)** Most of the primary auditory cortex is in the temporal lobe, but the association auditory cortices extend over much of the insular lobe and even onto the lateral portion of the parietal lobe.

TMP13 p. 680

- 68. B)** The termination of taste fibers for all taste sensations is in the nucleus of the solitary tract in the medulla.

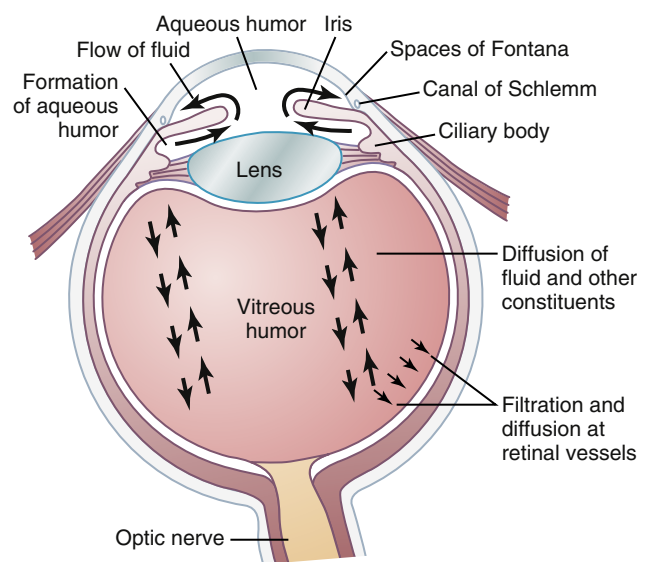
TMP13 pp. 687-688

- 69. B)** cGMP-dependent sodium channels in the outer portions of the rods and cones allow sodium ions to pass from the extracellular space to the intracellular space of the photoreceptor. This process results in a membrane potential that is somewhat lower than the resting membrane potential of a typical neuron. The movement of the sodium ions and resulting electrical potential change as a result of this enhanced permeability is known as the *dark current*.

TMP13 p. 650

- 70. B)** The medial longitudinal fasciculus is a pathway for nerve fibers entering and leaving the oculomotor, trochlear, and abducens nuclei of the brain stem, thus allowing communication to coordinate the contraction of the various extraocular eye muscles.

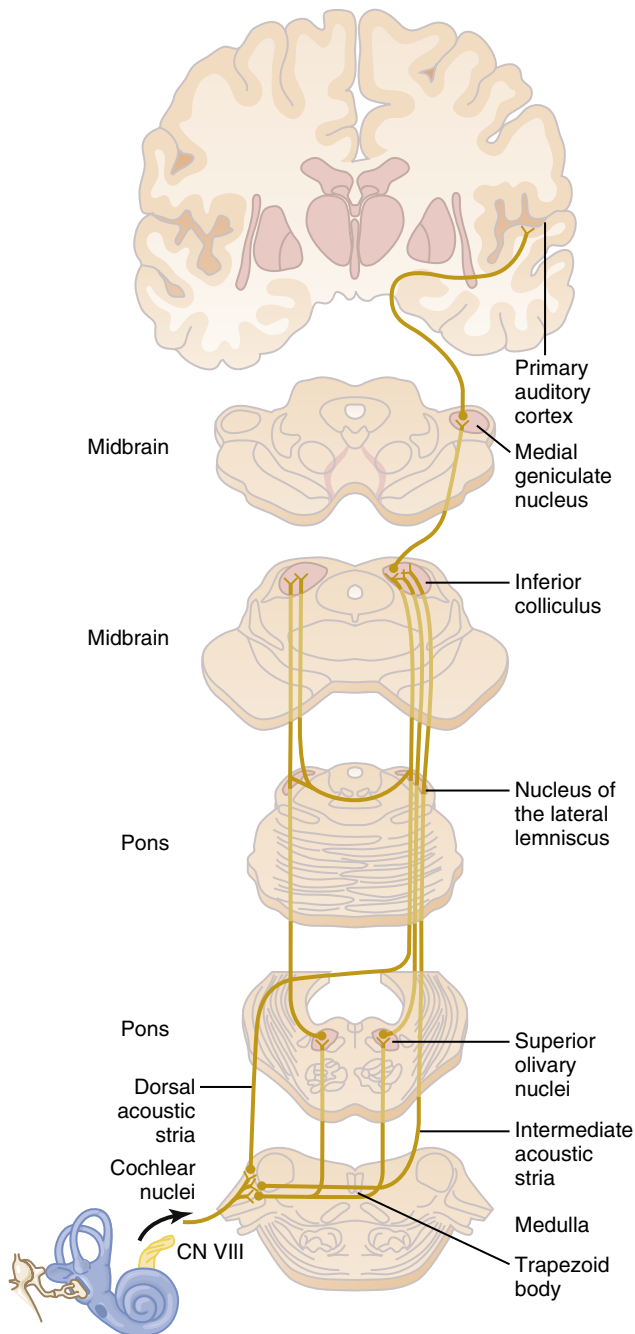
TMP13 p. 666



- 71. B)** Intraocular fluid flows from the anterior chamber of the eye, between the cornea and the iris through a meshwork of trabeculae into the canal of Schlemm, which empties into extraocular aqueous veins (see the figure above).

TMP13 p. 645

72. **C)** Only ganglion cells have action potentials. Photoreceptors, bipolar cells, amacrine cells, and horizontal cells all appear to operate through graded potentials.
TMP13 p. 655



73. **D)** The superior olivary nuclei (see the figure at left) receive auditory information from both ears and begin the process of detecting the direction from which a sound comes. The lateral part of the superior olivary nucleus does so by comparing the difference in intensities of sound reaching the two ears, whereas the medial part of the superior olivary nucleus detects time lag between signals entering both ears.

TMP13 pp. 681-682

74. **D)** The outputs of horizontal cells are always inhibitory; their lateral connections with synaptic bodies of photoreceptors (rods and cones) and dendrites of bipolar cells provide lateral inhibition to ensure transmission of visual patterns with proper visual contrast. Lateral inhibition is critical in all sensory systems to sharpen the sensory signals. There are many types of amacrine cells with at least six types of functions; they transmit signals both horizontally and vertically, forming connections with many different cell types. Bipolar cells transmit signals vertically from photoreceptors and horizontal cells to ganglion cells and amacrine cells in the inner plexiform layer of the retina. Ganglion cells transmit output signals from the retina through the optic nerve to the brain.

TMP13 pp. 656-657

75. **A)** Foliate papillae are located in the folds along the lateral surfaces of the tongue, fungiform papillae are located in the anterior part of the tongue, and circumvallate papillae are located in the posterior part of the tongue. The papilla of Vater empties pancreatic secretions and bile into the duodenum.

TMP13 p. 687

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The Nervous System: C. Motor and Integrative Neurophysiology

1. A 76-year-old man has a stroke that severely impairs his speech. Which area of his brain is most likely damaged?
 - A) Primary motor cortex
 - B) Premotor area
 - C) Broca's area
 - D) Cerebellum
 2. A 17-year-old boy sustains serious head and neck trauma during a football game. Physical examination shows a positive Babinski sign. What part of the brain has most likely been damaged in this boy?
 - A) Anterior motor neurons
 - B) Cerebellum
 - C) Corticospinal tract
 - D) Premotor cortex
 3. Which statement best describes a functional role for the lateral hemispheres of the cerebellum?
 - A) Control and coordinate movements of the axial muscles, as well as the shoulder and hip
 - B) Control movements that involve distal limb musculature
 - C) Function with the cerebral cortex to plan movements
 - D) Stimulate motor neurons through their connections to the spinal cord
 4. In which type of neuron does the axon form synaptic junctions with the skeletal muscle cells (extrafusal fibers) that comprise the major part of a muscle?
 - A) Alpha motor neuron
 - B) Pyramidal neuron
 - C) Gamma motor neuron
 - D) Granule cell
 - E) Purkinje cell
 5. Which of the following would produce an increase in cerebral blood flow?
 - A) Increase in carbon dioxide concentration
 - B) Increase in oxygen concentration
 - C) Decrease in the activity of cerebral cortex neurons
 - D) Decrease in carbon dioxide concentration
 - E) Decrease in arterial blood pressure from 120 mm Hg to 90 mm Hg
 6. As the axons of motor neurons leave the spinal cord and course peripherally to skeletal muscle, they must pass through which structure?
 - A) Posterior column
 - B) Posterior root
 - C) Ventral white commissure
 - D) Posterior horn
 - E) Anterior root
 7. Which spinal cord level contains the entire population of preganglionic sympathetic neurons?
 - A) C5-T1
 - B) C3-C5
 - C) S2-S4
 - D) T1-L2
 - E) T6-L1
- Questions 8 and 9**
- A left-side subdural hematoma develops in a 23-year-old man after an automobile accident. Physical examination shows papilledema 3 days after the accident. Use this information to answer Questions 8 and 9.
8. Which of the following is most likely to be increased in this patient?
 - A) Cerebral blood flow
 - B) Cerebrospinal fluid production
 - C) Cerebrospinal fluid volume
 - D) Intracranial pressure
 - E) Intracranial venous volume
 9. Collapse of which of the following structures is most likely to lead to a decrease in brain oxygenation in this patient?
 - A) Arteries
 - B) Capillaries
 - C) Lateral ventricles
 - D) Subarachnoid space
 - E) Veins
 10. Preganglionic sympathetic axons pass through which of the following structures?
 - A) Dorsal root
 - B) Dorsal primary rami
 - C) White rami
 - D) Gray rami
 - E) Ventral primary rami

11. Which statement best describes a functional role for the intermediate zone of the cerebellum?
 - A) Controls and coordinates movements of the axial muscles, as well as the shoulder and hip
 - B) Controls movements that involve distal limb musculature
 - C) Functions with the cerebral cortex to plan movements
 - D) Stimulates motor neurons through its connections to the spinal cord
12. Which body part is represented most laterally and inferiorly within the primary motor cortex?
 - A) Face
 - B) Hand
 - C) Neck
 - D) Abdomen
 - E) Lower limb
13. The gigantocellular neurons of the reticular formation release which neurotransmitter?
 - A) Norepinephrine
 - B) Serotonin
 - C) Dopamine
 - D) Acetylcholine
 - E) Glutamate
14. Astrocytes participating in the metabolic control of cerebral blood flow have the following three events associated with the process: (1) prostaglandin release, (2) a calcium wave, and (3) glutamate spillover. Which sequence best describes the correct temporal order of these three events?
 - A) 2, 1, 3
 - B) 1, 2, 3
 - C) 3, 1, 2
 - D) 1, 3, 2
 - E) 3, 2, 1
 - F) 2, 3, 1
15. A 15-year-old girl is taken to see a physician because of a sore throat. An antibiotic is prescribed that can enter most tissues of the body but cannot penetrate the blood-brain barrier. The blood-brain barrier can be attributed primarily to which cell type?
 - A) Astrocyte
 - B) Endothelial cell
 - C) Glial cell
 - D) Macrophage
 - E) Pericyte
 - F) Smooth muscle cell
16. In which type of neuron does the axon form synaptic junctions with skeletal muscle cells (intrafusal fibers) within the muscle spindles?
 - A) Alpha motor neuron
 - B) Pyramidal neuron
 - C) Gamma motor neuron
 - D) Granule cell
 - E) Purkinje cell

17. Which statement best describes a functional role for the cerebellar vermis?
 - A) Controls and coordinates movements of the axial muscles, as well as the shoulder and hip
 - B) Controls movements that involve distal limb musculature
 - C) Functions with the cerebral cortex to plan movements
 - D) Stimulates motor neurons through its connections to the spinal cord
18. Which projection system is contained in the superior cerebellar peduncle?
 - A) Pontocerebellar
 - B) Cerebellothalamic
 - C) Posterior spinocerebellar
 - D) Corticospinal

Questions 19 and 20

A 29-year-old man steps on a broken bottle with his bare right foot. His right leg immediately lifts while his left leg extends before he can consciously react to the pain. Use this information to answer Questions 19 and 20.

19. This action is attributable to which reflex?
 - A) Walking reflex
 - B) Stretch reflex
 - C) Patellar tendon reflex
 - D) Golgi tendon reflex
 - E) Flexor withdrawal reflex
20. Which of the following best describes the type of reflex arc and sensory receptor for this reflex?

	Reflex Arc	Sensory Receptor
A)	Disynaptic	Pacian corpuscle
B)	Disynaptic	Nociceptor
C)	Monosynaptic	Pacian corpuscle
D)	Monosynaptic	Golgi tendon organ
E)	Polysynaptic	Nociceptor
F)	Polysynaptic	Muscle spindle
21. Which brain structure serves as the major controller of the limbic system?
 - A) Hypothalamus
 - B) Hippocampus
 - C) Amygdala
 - D) Mammillary body
 - E) Fornix
22. A large portion of the cerebral cortex does not fit into the conventional definition of motor or sensory cortex. Which term refers to the type of cortex that receives input primarily from several other regions of the cerebral cortex?
 - A) Cortex that is agranular
 - B) Secondary somatosensory cortex
 - C) Association cortex
 - D) Supplementary motor cortex
 - E) Secondary visual cortex

23. The two hemispheres of the brain are connected by which nerve fibers or pathways?
 - A) Lateral lemniscus
 - B) Corticofugal fibers
 - C) Corpus callosum
 - D) Arcuate fasciculus
 - E) Medial longitudinal fasciculus
24. The fibers of the corticospinal tract pass through which structure?
 - A) Medial lemniscus
 - B) Medullary pyramid
 - C) Posterior funiculus
 - D) Medial longitudinal fasciculus
 - E) Anterior roots
25. The condition of prosopagnosia usually results from dysfunction or damage to which area of the cerebral cortex?
 - A) Prefrontal area
 - B) Junction of the parietal and temporal lobe on the nondominant side of the brain
 - C) Frontal eye fields
 - D) Underside of the medial occipital and temporal lobes
 - E) Limbic association areas of frontal and anterior temporal lobes
26. Lesions of which area of the brain would have the most devastating effect on verbal and symbolic intelligence?
 - A) Hippocampus
 - B) Amygdala
 - C) Wernicke's area on the nondominant side of the brain
 - D) Broca's area
 - E) Wernicke's area on the dominant side of the brain
27. Which term applies to the combination of a motor neuron and all the skeletal muscle fibers contacted by that motor neuron?
 - A) Golgi tendon organ
 - B) Motor unit
 - C) Propriospinal neurons
 - D) Skeletal muscle fibers
28. Which maneuver will attenuate the stretch reflex in skeletal muscle?
 - A) Sectioning the dorsal root of a spinal nerve
 - B) Disruption of the spinocerebellar tract
 - C) Disruption of the corticospinal tract
 - D) Sectioning the medial lemniscus on the contralateral side of the skeletal muscle in question
 - E) Creating a lesion in the contralateral globus pallidus
29. A stroke involving the middle cerebral artery on the left side is likely to cause which symptom?
 - A) Paralysis of the left side of the face and left upper extremity
 - B) Paralysis of left lower extremity
 - C) Complete loss of vision in both eyes
 - D) Loss of ability to comprehend speech
 - E) Loss of vision in the left half of both eyes
30. The creation of memory can be interrupted by which activity?
 - A) Phosphorylation of a potassium channel to block activity
 - B) Activation of adenylate cyclase
 - C) Unnatural loss of consciousness
 - D) Increase in protein synthesis
 - E) Activation of cyclic guanosine monophosphate (cGMP) phosphodiesterase
31. Which structure serves to connect Wernicke's area to Broca's area in the cerebral cortex?
 - A) Arcuate fasciculus
 - B) Lateral lemniscus
 - C) Medial longitudinal fasciculus
 - D) Anterior commissure
 - E) Internal capsule
32. Broca's area is a specialized portion of motor cortex. Which condition best describes the deficit resulting from damage to Broca's area?
 - A) Spastic paralysis of the contralateral hand
 - B) Paralysis of the muscles of the larynx and pharynx
 - C) Inability to use two hands to grasp an object
 - D) Inability to direct the two eyes to the contralateral side
 - E) Inability to speak whole words correctly
33. Which projection system is contained in the inferior cerebellar peduncle?
 - A) Pontocerebellar
 - B) Cerebellothalamic
 - C) Posterior spinocerebellar
 - D) Corticospinal
 - E) Dorsospinocerebellar
34. Signals from motor areas of the cortex reach the contralateral cerebellum after first passing through which structure?
 - A) Thalamus
 - B) Caudate nucleus
 - C) Red nucleus
 - D) Basilar pontine nuclei
 - E) Dorsal column nuclei
35. Cerebrospinal fluid (CSF) provides a cushioning effect both inside and outside the brain. Which space that lies outside the brain or spinal cord contains CSF?
 - A) Lateral ventricle
 - B) Third ventricle
 - C) Cisterna magna
 - D) Epidural space
 - E) Aqueduct of Sylvius

Questions 36 and 37

A 40-year-old woman visits the physician because of uncontrolled movements of her arms, legs, head, face, and upper body. These symptoms have increased progressively during the past 12 months. She is also depressed and irritable, and she repeats the same question six times during the 30-minute office visit. Gene analyses show expansion of a CAG triplet repeat on chromosome 4. Use this information to answer Questions 36 and 37.

36. Which diagnosis is most likely?
 - A) Alzheimer's disease
 - B) Bipolar disorder
 - C) Brain tumor
 - D) Huntington's disease
 - E) Parkinson's disease
37. Which of the following is most likely to be decreased in this woman?
 - A) Acetylcholine neurons in the magnocellular fore-brain nucleus
 - B) Dopamine neurons in the substantia nigra
 - C) γ -Aminobutyric acid (GABA) neurons in the caudate nucleus and putamen
 - D) Serotonin neurons in the raphe nuclei
38. Which projection system is contained in the middle cerebellar peduncle?
 - A) Pontocerebellar
 - B) Cerebellothalamic
 - C) Posterior spinocerebellar
 - D) Corticospinal
 - E) Ventrospinocerebellar
39. The peripheral sensory input that activates the ascending excitatory elements of the reticular formation comes mainly from which of the following?
 - A) Pain signals
 - B) Proprioceptive sensory information
 - C) Corticospinal system
 - D) Medial lemniscus
 - E) Input from Pacinian corpuscles
40. Cells of the adrenal medulla receive synaptic input from which type of neuron?
 - A) Preganglionic sympathetic
 - B) Postganglionic sympathetic
 - C) Preganglionic parasympathetic
 - D) Postsynaptic parasympathetic
 - E) Presynaptic parasympathetic
41. Which activity will increase the sensitivity of the stretch reflex?
 - A) Cutting the dorsal root fibers associated with the muscle in which the stretch reflex is being examined
 - B) Increasing the activity of the medullary reticular nuclei
 - C) Bending the head forward
 - D) Enhanced activity in the fusimotor (gamma motor neuron) system
 - E) Stimulating the lateral hemispheres of the cerebellum
42. Neurological disease associated with the cerebellum produces which type of symptoms?
 - A) Resting tremor
 - B) Athetosis
 - C) Rigidity
 - D) Ataxia
 - E) Akinesia
43. Preganglionic parasympathetic neurons that contribute to the innervation of the descending colon and rectum are found in which structure?
 - A) Superior cervical ganglion
 - B) Dorsal motor nucleus of the vagus
 - C) Superior mesenteric ganglion
 - D) Ciliary ganglion
 - E) Spinal cord levels S2 and S3
44. A complex spike pattern in the Purkinje cells of the cerebellum can be initiated by stimulation of which brain area?
 - A) Inferior olivary complex
 - B) Brain stem reticular nuclei
 - C) Neurons in red nucleus
 - D) Superior olivary complex
 - E) Dorsal vestibular nucleus
45. In a muscle spindle receptor, which type of muscle fiber is responsible for the dynamic response?
 - A) Extrafusal muscle fiber
 - B) Static nuclear bag fiber
 - C) Nuclear chain fiber
 - D) Dynamic nuclear bag fiber
 - E) Smooth muscle fiber
46. Which structure serves as an "alternative pathway" for signals from the motor cortex to the spinal cord?
 - A) Red nucleus
 - B) Basilar pontine nuclei
 - C) Caudate nucleus
 - D) Thalamus
 - E) Dorsal column nuclei
47. The phenomenon of decerebrate rigidity can be explained, at least in part, by which of the following?
 - A) Stimulation of type 1b sensory neurons
 - B) Loss of cerebellar inputs to the red nucleus
 - C) Overactivity of the medullary reticular nuclei involved in motor control
 - D) Unopposed activity of the pontine reticular nuclei
 - E) Degeneration of the nigrostriatal pathway

48. Like the primary visual cortex, the primary motor cortex is organized into vertical columns composed of cells linked together throughout the six layers of the cortex. The cells that contribute axons to the corticospinal tract are concentrated in which cortical layer?

- A) Layer I
- B) Layer II
- C) Layer III
- D) Layer IV
- E) Layer V

Questions 49 and 50

A 60-year-old man is taken to the physician because of a tremor in his hands, trouble sleeping, constipation, and dizziness. Physical examination shows a resting tremor, rigidity, and bradykinesia. The man is alert, engaging, and optimistic. He speaks in a low, soft voice. Use this information to answer Questions 49 and 50.

49. Which diagnosis is most likely?

- A) Alzheimer's disease
- B) Bipolar disorder
- C) Brain tumor
- D) Huntington's disease
- E) Parkinson's disease

50. Which of the following is most likely to be decreased in this man?

- A) Serotonin neurons in the raphe nuclei
- B) GABA neurons in the caudate nucleus and putamen
- C) Dopamine neurons in the substantia nigra
- D) Acetylcholine neurons in the magnocellular fore-brain nucleus

51. Motor cortex neurons receive feedback from muscles activated by the corticospinal system. This feedback arises from which of the following structures?

- A) Red nucleus
- B) Spinocerebellar tracts
- C) Skin surface of fingers used to grasp an object
- D) Muscle spindles in muscles antagonistic to those used to make the movement
- E) Vestibular nuclei

52. The sweat glands and piloerector muscles of hairy skin are innervated by which type of fibers?

- A) Cholinergic postganglionic parasympathetic
- B) Cholinergic postganglionic sympathetic
- C) Adrenergic preganglionic parasympathetic
- D) Adrenergic postganglionic sympathetic
- E) Adrenergic preganglionic sympathetic

53. In a neurophysiology experiment conducted with monkeys, the amygdalae are surgically ablated bilaterally. Which of the following is most likely to be increased 6 months after ablation of the amygdala?

- A) Despondence
- B) Memory
- C) Paranoia
- D) Sex drive
- E) Tremors

54. In controlling the fine muscles of the hands and fingers, corticospinal axons can synapse primarily with which of the following?

- A) Posterior horn neurons
- B) Spinal cord interneurons
- C) Spinal cord motor neurons
- D) Purkinje cells
- E) Renshaw cells

55. Which of the following foramina allows cerebrospinal fluid to pass directly from the ventricular system into the subarachnoid space?

- A) Foramen of Magendie
- B) Aqueduct of Sylvius
- C) Third ventricle
- D) Lateral ventricle
- E) Arachnoid villi

56. Which epileptic condition involves a postictal depression period lasting from several minutes to perhaps as long as several hours?

- A) Generalized tonic-clonic seizure
- B) Absence seizure
- C) Jacksonian seizure
- D) Phase-out clonic seizure
- E) Temporal lobe seizure

57. An area in the dominant hemisphere, when damaged, may leave the sense of hearing intact but not allow words to be arranged into a comprehensive thought. Which term is used to identify this portion of the cortex?

- A) Primary auditory cortex
- B) Wernicke's area
- C) Broca's area
- D) Angular gyrus
- E) Limbic association cortex

58. Afferent signals from the periphery of the body travel to the cerebellum in which nerve tract?

- A) Ventral spinocerebellar
- B) Fastigioreticular
- C) Vestibulocerebellar
- D) Reticulocerebellar
- E) Dorsal spinocerebellar

59. Which cells receive direct synaptic input from Golgi tendon organs?
- A) Type Ia inhibitory interneurons
 - B) Dynamic gamma motor neurons
 - C) Alpha motor neurons
 - D) Type Ib inhibitory interneurons**
 - E) Type II excitatory interneurons
60. Which neurotransmitter is used by the axons of locus coeruleus neurons that distribute throughout much of the brain?
- A) Norepinephrine**
 - B) Dopamine
 - C) Serotonin
 - D) Acetylcholine

Questions 61 and 62

A 45-year-old man visits the physician because of difficulties performing simple tasks that involve repetitive movements. The physician asks the patient to turn one hand upward and downward at a rapid pace. The man quickly loses all perception of the instantaneous position of the hand, which results in a series of stalled attempts and jumbled movements. Use this information to answer Questions 61 and 62.

61. Which term best describes this patient's movements?
- A) Agraphesthesia
 - B) Astereognosis
 - C) Dysarthria
 - D) Dysdiadochokinesia**
 - E) Hemineglect
62. Which area of his brain is most likely to have a lesion?
- A) Cerebellum**
 - B) Limbic system
 - C) Medulla oblongata
 - D) Premotor cortex
 - E) Primary motor cortex
63. The excitatory or inhibitory effect of a postganglionic sympathetic fiber is determined by which feature or structure?
- A) Function of the postsynaptic receptor to which it binds**
 - B) Specific organ innervated
 - C) Ganglion where the postganglionic fiber originates
 - D) Ganglion containing the preganglionic fiber
 - E) Emotional state of the individual
64. Which of the following correctly describes the relationship of CSF pressure to the venous pressure in the superior sagittal sinus?
- A) A few millimeters higher**
 - B) A few millimeters lower
 - C) Equal to
 - D) Twice the value
 - E) One-half the value
65. A vascular lesion that causes degeneration of corticospinal axons in the basilar pons will most likely lead to which condition?
- A) Paralysis primarily involving muscles around the contralateral shoulder and hip joints
 - B) Paralysis of the muscles of mastication
 - C) Loss of voluntary control of discrete movements of the contralateral hand and fingers**
 - D) Inability to speak clearly
 - E) Inability to convert short-term memory to long-term memory
66. Fine motor movement of the index finger can be elicited by stimulation of which brain area?
- A) Primary motor cortex**
 - B) Lateral cerebellar hemisphere
 - C) Premotor cortex
 - D) Supplemental motor area
 - E) Red nucleus
67. Which type of cholinergic receptor is found at synapses between preganglionic and postganglionic neurons of the sympathetic system?
- A) Muscarinic
 - B) Nicotinic
 - C) Alpha
 - D) Beta-1
 - E) Beta-2
68. A 23-year-old basketball player mentally rehearses free throw shots while lying in bed. Which option best describes the area of the brain that is involved in generating a motor image of this action in the absence of actual movement?
- A) Basal ganglia
 - B) Cerebellum
 - C) Limbic system
 - D) Premotor cortex**
 - E) Primary motor cortex
69. The perivascular space (Virchow-Robin space) in the brain is formed between the wall of small penetrating vessels and which structure?
- A) Dura mater
 - B) Arachnoid membrane
 - C) Pia mater**
 - D) Choroid plexus
 - E) Ependymal cells
70. Which type of seizure is associated with a spike and dome electroencephalogram pattern during the seizure activity?
- A) Generalized tonic-clonic
 - B) Temporal lobe
 - C) Jacksonian
 - D) Absence
 - E) Apoplectic

71. Which substance has the lowest concentration in the cerebrospinal fluid compared with the cerebral blood plasma?
- A) Chloride
 - B) Glucose
 - C) Potassium
 - D) Protein**
 - E) Sodium
72. The formation of cerebrospinal fluid by the choroid plexus includes (1) osmosis of water, (2) active transport of sodium, and (3) passive diffusion of chloride. Which sequence best describes the correct temporal order of these processes?
- A) 2, 3, 1**
 - B) 3, 2, 1
 - C) 1, 3, 2
 - D) 3, 1, 2
 - E) 1, 2, 3
 - F) 2, 1, 3
- Questions 73 and 74**
- A 10-year-old girl is taken to the physician because of difficulty walking. Physical examination shows loss of tendon reflexes in the knees and ankles and reduced two-point discrimination in the hands and feet. Repeat visits to the physician show a progressive worsening of these symptoms during the next 2 years. However, the girl is always alert and seems to have normal reasoning abilities. Her uncle had similar problems at age 12 years and later developed scoliosis followed by loss of hearing and vision. Use this information to answer Questions 73 and 74.
73. What is the most likely diagnosis?
- A) Friedreich's ataxia
 - B) Huntington's disease
 - C) Multiple sclerosis
 - D) Parkinson's disease
 - E) Poliomyelitis
74. What is the most likely cause of these symptoms in this girl?
- A) A lesion in the premotor cortex
 - B) A lesion in the primary motor cortex
 - C) Malformation of the cerebellum
 - D) Malformation of the frontal lobe
 - E) Nerve degeneration
 - F) Nerve proliferation
75. Which neurotransmitter is used by the axons of substantia nigra neurons that project to the caudate and putamen?
- A) Norepinephrine
 - B) Dopamine**
 - C) Serotonin
 - D) Acetylcholine
 - E) GABA
76. Damage limited to the primary motor cortex (area 4) is thought to cause hypotonia in the affected muscles. However, most cortical lesions, particularly those caused by vascular infarcts, generally involve the primary motor cortex in addition to surrounding areas of cortex or cortical efferent axons. The latter type of cortical lesion will cause which of the following?
- A) Spastic muscle paralysis**
 - B) Flaccid muscle paralysis
 - C) No paralysis—only jerky, fast movements
 - D) Complete blindness in the contralateral eye
 - E) Loss of sensation in the contralateral foot
77. The term *limbic cortex* includes the orbitofrontal cortex, subcallosal gyrus, cingulate gyrus, and which area?
- A) Supplementary motor cortex
 - B) Postcentral gyrus
 - C) Lingual gyrus
 - D) Parahippocampal gyrus**
 - E) Paracentral lobule
78. Which substance activates adrenergic alpha and beta receptors equally well?
- A) Acetylcholine
 - B) Norepinephrine
 - C) Epinephrine
 - D) Serotonin
 - E) Dopamine
79. The posterior and lateral hypothalamus, in combination with the preoptic area, are involved in the control of which of the following functions?
- A) Cardiovascular functions involving blood pressure and heart rate**
 - B) Regulation of thirst and water intake
 - C) Stimulation of uterine contractility and milk ejection from the breast
 - D) Signaling that food intake is sufficient (satiety)
 - E) Secretion of hormones from the anterior lobe of the pituitary gland
80. In the patellar tendon reflex, which of the following items will synapse directly on alpha motor neurons that innervate the muscle being stretched?
- A) Ia sensory fiber**
 - B) Ib sensory fiber
 - C) Excitatory interneurons
 - D) Gamma motor neurons
 - E) Inhibitory interneurons
81. Occlusion of which structure would lead to communicating hydrocephalus?
- A) Aqueduct of Sylvius
 - B) Lateral ventricle
 - C) Foramen of Luschka
 - D) Foramen of Magendie
 - E) Arachnoid villi**

82. Evaluation of a patient reveals the following deficits: (1) decreased aggressiveness and ambition and inappropriate social responses; (2) inability to process sequential thoughts in order to solve a problem; and (3) inability to process multiple bits of information that could then be recalled instantaneously to complete a thought or solve a problem. Damage to which brain region could be responsible for such deficits?
- Premotor cortex
 - Parieto-occipital cortex in the nondominant hemisphere
 - Broca's area
 - Limbic association cortex
 - Prefrontal association cortex**
83. A lesion in Wernicke's cortical area in the dominant hemisphere is most likely to produce which symptoms?
- Impaired language skills**
 - Impaired motor skills
 - Inability to form new memories
 - Inability to plan future movements
 - Reduced cerebellar activity
 - Reduced cerebral cortex activity
84. Which of the following represents the structural basis of the blood-CSF barrier?
- Tight junctions between the ependymal cells forming the ventricular walls
 - Arachnoid villi
 - Tight junctions between adjacent choroid plexus cells**
 - Astrocyte foot processes
 - Tight junctions between adjacent endothelial cells of brain capillaries
85. The withdrawal reflex is initiated by stimulation delivered to which receptor?
- Muscle spindle
 - Joint capsule receptor
 - Cutaneous free nerve ending**
 - Golgi tendon organ
 - Pacinian corpuscle
86. A 21-year-old woman is a right-handed musician of considerable talent. Which brain structure is most likely to have been physically larger in the dominant hemisphere compared with the nondominant hemisphere at birth?
- Anterior temporal lobe
 - Posterior temporal lobe**
 - Premotor cortex
 - Primary motor cortex
 - Primary somatosensory area
 - Sensory association area
87. Nasal, lacrimal, salivary, and gastrointestinal glands are stimulated by which substance?
- Acetylcholine
 - Norepinephrine
 - Epinephrine
 - Serotonin
 - Dopamine
88. The neurons located in the locus coeruleus release which neurotransmitter at their synaptic terminals?
- Norepinephrine**
 - Dopamine
 - GABA
 - Acetylcholine
 - Serotonin
89. Which of the following reflexes best describes incoming pain signals that elicit movements performed by antagonistic muscle groups on either side of the body?
- Crossed extensor reflex**
 - Withdrawal reflex
 - Reciprocal inhibition
 - Autogenic inhibition
90. Which portion of the cerebellum functions in the planning of sequential movement?
- Vermis and fastigial nucleus
 - Intermediate zone and fastigial nucleus
 - Lateral hemisphere and interposed nucleus
 - Cerebrocerebellum and dentate nucleus**
 - Spinocerebellum and interposed nucleus
91. Which reflex is correctly paired with the sensory structure that mediates the reflex?
- Autogenic inhibition—muscle spindle
 - Reciprocal inhibition—Golgi tendon organ
 - Reciprocal inhibition—Pacinian corpuscle
 - Stretch reflex—muscle spindle**
 - Golgi tendon reflex—Meissner corpuscle
92. Damage to which brain area leads to the inability to comprehend the written or the spoken word?
- Insular cortex on the dominant side of the brain
 - Anterior occipital lobe
 - Junction of the parietal, temporal, and occipital lobes**
 - Medial portion of the precentral gyrus
 - Most anterior portion of the temporal lobe
93. A computed tomography scan of a newborn boy shows agenesis of the corpus callosum. Which of the following is most likely to occur in this child during the next 5 years as he matures?
- Inability to form new memories
 - Inability to understand spoken words
 - Inability to verbally express words
 - Reduction in communication between the two hemispheres**
 - Tameness and inability to recognize expressions of fear

94. A 67-year-old man has a stroke. One week later, he experiences sudden and uncontrolled flailing, ballistic movements of his limbs. Which part of the man's brain is most likely to have been damaged by the stroke?
- Globus pallidus
 - Lateral hypothalamus
 - Red nucleus
 - Subthalamic nucleus**
 - Ventrobasal complex of thalamus
95. A physiology experiment is conducted in which a test dose of norepinephrine is administered intravenously to the front limbs of rats, causing a 25 percent reduction in blood flow to the front limbs compared with basal values. Next, the stellate ganglion is removed. Three days later the same dose of norepinephrine is administered intravenously. Which option best describes the most likely change in front limb blood flow (compared with basal values) within 30 minutes after norepinephrine is administered to the ganglionectomized rats?
- 25 percent increase
 - 25 percent reduction
 - 5 percent increase
 - 5 percent reduction
 - 75 percent increase
 - 75 percent reduction
96. In an otherwise normal person, dysfunction of which brain area will lead to behavior that is not appropriate for the given social occasion?
- Ventromedial nuclei of hypothalamus
 - Amygdala**
 - Corpus callosum
 - Fornix
 - Uncus
97. The function of which organ or system is dominated by the sympathetic nervous system?
- Systemic blood vessels**
 - Heart
 - Gastrointestinal gland secretion
 - Salivary glands
 - Gastrointestinal motility
98. Schizophrenia is thought to be caused in part by excessive production and release of which neurotransmitter agent?
- Norepinephrine
 - Serotonin
 - Acetylcholine
 - Substance P
 - Dopamine**
99. Stimulation of which subcortical area can lead to contraction of a single muscle or small groups of muscles?
- Dentate nucleus of the cerebellum
 - Ventrobasal complex of the thalamus
 - Red nucleus**
 - Subthalamic nucleus
 - Nucleus accumbens
100. Bilateral lesions involving the ventromedial hypothalamus will lead to which of the following deficits?
- Decreased eating and drinking
 - Loss of sexual drive
 - Excessive eating, rage and aggression, hyperactivity**
 - Uterine contractility, mammary gland enlargement
 - Obsessive compulsive disorder
101. Under awake, resting conditions, brain metabolism accounts for about 15 percent of the total metabolism of the body; this rate is among the highest metabolic rates of all tissues in the body. Which cellular population of the nervous system contributes most substantially to this high rate of metabolism?
- Astrocytes
 - Neurons**
 - Ependymal cells
 - Choroid plexus cells
 - Brain endothelial cells
102. Which structure(s) in the cerebellum has/have a topographical representation of the body?
- Dentate nucleus
 - Lateral hemispheres
 - Flocculonodular lobe
 - Vermis and intermediate hemisphere**
 - Cerebellar peduncle
103. Which structure is an important pathway for communication between the limbic system and the brain stem?
- Mammillothalamic tract
 - Fornix
 - Anterior commissure
 - Indusium griseum
 - Medial forebrain bundle
104. A 75-year-old woman is taken to the physician because of worsening forgetfulness. She has trouble playing cards with her friends because she cannot remember what game is being played. She recently got lost during a walk in the neighborhood she has lived in for 35 years. Which substance is most likely to be increased in the brain of this woman?
- alpha-1 antitrypsin
 - alpha-amylase
 - beta-amyloid peptide
 - beta-endorphin
 - gamma-glutamyl hydrolase
 - gamma-glutamyl transferase
105. Which of the following best describes the cerebellar deficit in which there is a failure to perform rapid alternating movements indicating a failure of "progression" from one part of the movement to the next?
- Past-pointing
 - Intention tremor
 - Dysarthria
 - Cerebellar nystagmus
 - Dysdiadochokinesia**

106. Which structure in the vestibular apparatus is responsible for the detection of angular acceleration?
- Statoconia
 - Macula
 - Semicircular canals
 - Sacculae
 - Ampullae
107. The concept of “autonomic tone” is quite advantageous because it allows the nervous system to have much finer control over the function of an organ or organ system than would otherwise be possible. This ability is exemplified in the control of systemic arterioles. Which action would lead to vasodilation of systemic arterioles?
- Increased activity of preganglionic parasympathetic neurons
 - Decreased activity of postganglionic parasympathetic neurons
 - Increased activity of postganglionic sympathetic neurons
 - Decreased activity of postganglionic sympathetic neurons
 - Increased activity of preganglionic sympathetic neurons
108. A person who has had a traumatic brain injury seems to be able to understand the written and spoken word but cannot create the correct sounds to be able to speak a word that is recognizable. This person most likely has damage to which area of the brain?
- Wernicke’s area
 - Broca’s area
 - Angular gyrus
 - Dentate nucleus
 - Prefrontal lobe
111. Which structure is maximally sensitive to linear head movement in the vertical plane?
- Macula of the utricle
 - Macula of the saccule
 - Crista ampullaris of the anterior semicircular duct
 - Crista ampullaris of the horizontal semicircular duct
112. Retrograde amnesia is the inability to recall long-term memories. Damage to which brain region leads to retrograde amnesia?
- Hippocampus
 - Dentate gyrus
 - Amygdaloid complex
 - Thalamus
 - Mammillary nuclei of hypothalamus
113. Which component of the basal ganglia plays the major role in the control of cognitive (memory-guided) motor activity?
- Globus pallidus
 - Substantia nigra
 - Caudate nucleus
 - Putamen
 - Subthalamic nucleus
114. A 9-month-old boy is brought to the emergency department by his grandmother because of irritability and vomiting. The parents dropped the baby off at the grandmother’s house 1 hour ago; their current whereabouts are not known. Magnetic resonance imaging shows retinal hemorrhages in both eyes, a subdural hematoma, and cerebral edema. Which of the following is most likely to be increased in this infant?
- Brain oxygenation
 - Cerebral venous volume
 - Intracranial pressure
 - Visual acuity
115. Stimulation of the punishment center can inhibit the reward center, demonstrating that fear and punishment can take precedence over pleasure and reward. Which of the following cell groups is considered the punishment center?
- Lateral and ventromedial hypothalamic nuclei
 - Periventricular hypothalamus and midbrain central gray area
 - Supraoptic nuclei of hypothalamus
 - Anterior hypothalamic nucleus
116. Drugs that stimulate specific adrenergic receptors are called *sympathomimetic drugs*. Which drug is a sympathomimetic drug?
- Reserpine
 - Phentolamine
 - Propranolol
 - L-dopa
 - Phenylephrine

Questions 109 and 110

A 45-year-old man is taken to the psychiatrist because of delusional behavior in the workplace. The man accused a co-worker of scheming with his neighbor to transplant poison ivy in his backyard. This plot was revealed to the man by a voice in his head. Other examples of delusional thinking and voices in the man’s head are abundant. Use this information to answer Questions 109 and 110.

109. What is the most likely diagnosis?
- Bipolar disorder
 - Dissociative identity disorder
 - Multiple personality disorder
 - Schizophrenia
110. A decrease in size of which brain structure is most likely in this man?
- Globus pallidus
 - Hippocampus
 - Lateral hypothalamus
 - Red nucleus
 - Subthalamic nucleus

117. Although the sympathetic nervous system is often activated in such a way that it leads to mass activation of sympathetic responses throughout the body, it can also be activated to produce relatively discrete responses. Which option is an example of a local or discrete sympathetic action?
- A) Heating of a patch of skin causes a relatively restricted vasodilation in the heated region
 - B) Food in the mouth causes salivation
 - C) Emptying of the bladder may cause reflexive emptying of the bowel
 - D) Dust particles in the eye cause increased tear fluid release
 - E) Bright light introduced into one eye
118. An experimental drug is administered intravenously to six healthy volunteers. A unanimous finding in all six volunteers is decreased induction of sleep. A decrease in production of which substance is most likely in these volunteers after treatment with the experimental drug?
- A) Acetylcholine
 - B) Dopamine
 - C) Glutamate
 - D) Norepinephrine
 - E) Serotonin
119. A 10-year-old boy jumps off the porch and lands on the balls of his feet. The increase in muscle tension causes a sudden, complete relaxation of the affected muscles. Which sensory receptor is most likely to mediate this relaxation of muscles when tension is increased?
- A) Free nerve ending
 - B) Golgi tendon organ
 - C) Krause corpuscle
 - D) Muscle spindle
 - E) Pacinian corpuscle
120. Which structure connects the hippocampus to the limbic system?
- A) Mamillothalamic tract
 - B) Fornix
 - C) Anterior commissure
 - D) Medial forebrain bundle
 - E) Arcuate fasciculus
121. A wide variety of neurotransmitters have been identified in the cell bodies and afferent synaptic terminals in the basal ganglia. A deficiency of which transmitter is typically associated with Parkinson's disease?
- A) Norepinephrine
 - B) Dopamine
 - C) Serotonin
 - D) GABA
 - E) Substance P
122. The condition of athetosis results when which area of the brain is dysfunctional?
- A) Globus pallidus
 - B) Substantia nigra
 - C) Ventral anterior complex of the thalamus
 - D) Putamen
 - E) Purkinje cell layer of the cerebellum

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1. **C)** Broca's area is a region of the premotor area of one hemisphere (usually the left). Damage to Broca's area does not prevent a person from vocalizing but makes it impossible to speak whole words other than occasional simple words such as "yes" or "no." The primary motor cortex works with other areas of the brain to plan and execute movements. The cerebellum plays a critical role in motor control; it does not initiate movement but contributes to coordination, precision, and accurate timing of movements.
TMP13 pp. 708, 709
2. **C)** A positive Babinski sign (also called the *Babinski reflex*) occurs normally in children up to 2 years of age. The reflex occurs after the sole of the foot has been stroked with a blunt instrument; the big toe moves upward and the other toes fan out. A positive Babinski sign in adults can indicate damage to the corticospinal tract.
TMP13 pp. 705-706
3. **C)** The lateral cerebellar hemispheres function with the cerebral cortex in the planning of complex movements.
TMP13 p. 722
4. **A)** Alpha motor neurons form direct synaptic contact with skeletal extrafusal muscle fibers, whereas gamma motor neurons form synaptic junctions with intrafusal muscle fibers. Pyramidal, granule, and Purkinje neurons are located in the central nervous system and have no direct contact with skeletal muscle.
TMP13 p. 696
5. **A)** The most potent stimulator of cerebral blood flow is a local increase in carbon dioxide concentration, followed in order by a decrease in oxygen concentration and an increase in local neuronal activity.
TMP13 p. 788
6. **E)** Axons of motor neurons in the anterior horn exit the spinal cord through the anterior root. The posterior root serves as the entry point for sensory fibers coming into the posterior horn region of the spinal cord. The posterior column and ventral white commissure are fiber tracts located solely within the spinal cord.
TMP13 pp. 695-696
7. **D)** All preganglionic sympathetic neurons are located in the intermediolateral cell column (lateral horn); this cell group extends from T1 to L2.
TMP13 p. 774
8. **D)** A subdural hematoma can lead to increased intracranial pressure because it takes up space in the cranium; papilledema (optic disc swelling) suggests an increase in intracranial pressure. The increase in intracranial pressure does not affect production of CSF, but it may cause decreased CSF volume because the high pressure pushes CSF into venous blood through the arachnoidal villi and also compresses the volume of brain structures that contain CSF. Cerebral blood flow should remain normal with small increases in intracranial pressure, but larger increases can decrease cerebral blood flow.
TMP13 pp. 792-793
9. **E)** The veins have lower pressures compared with arteries and capillaries, making them easier to compress. When the veins are compressed, the capillary pressure increases, which increases the ultrafiltration of fluid from the capillaries into the interstitial spaces, thereby increasing the intracranial pressure even more. The increase in intracranial pressure can cause compression of lateral ventricles and the subarachnoid space, but this mechanism is compensatory rather than a cause for deterioration of blood flow and brain oxygenation.
TMP13 p. 793
10. **C)** Preganglionic sympathetic axons pass through the white communicating rami to enter the sympathetic trunk. Postganglionic sympathetic axons course through gray rami and might be found in dorsal and ventral primary rami.
TMP13 p. 773
11. **B)** The intermediate zone of the cerebellum influences the function of distal limb muscles.
TMP13 p. 722
12. **A)** The face region of the motor cortex is most inferior and lateral in the territory of the middle cerebral artery, whereas the lower limb is in the paracentral lobule in the territory of the anterior cerebral artery.
TMP13 p. 708
13. **D)** The gigantocellular neurons of the reticular formation reside in the pons and mesencephalon. These neurons release acetylcholine, which functions as an excitatory neurotransmitter in most brain areas.
TMP13 p. 753
14. **E)** Increased neuronal activity in the brain causes the neurotransmitter glutamate to diffuse from the site of release at the synapses into the adjacent tissues. The

glutamate triggers a calcium wave in astrocytes, which leads to astrocytic release of vasodilatory prostaglandins that cause arterioles to dilate. In this way, the local blood flow to the tissues can be matched with the metabolic activity of the neurons.

TMP13 p. 788

15. **B)** The endothelial cells lining all blood vessels in the brain constitute the blood-brain barrier. The purpose of the blood-brain barrier is to protect the chemical environment of the brain from rapid changes in composition that occur normally in the rest of the body fluids. Brain capillary endothelial cells have special structural and biochemical attributes that impede diffusion of ions, nutrients, and fat-soluble substances; these substances can diffuse through the endothelial barrier and thereby enter into all other tissues of the body.

TMP13 p. 793

16. **C)** Gamma motor neurons form direct synaptic contact with the skeletal muscle fibers known as *intrafusal fibers*. Extrafusal muscle fibers are innervated by alpha motor neurons, whereas Purkinje, granule, and pyramidal neurons have no synaptic contact with muscles in the periphery.

TMP13 p. 696

17. **A)** The cerebellar vermis is involved with the control of axial muscles and proximal limb muscles in the shoulder and hip.

TMP13 p. 722

18. **B)** Cerebellothalamic projections are contained in the superior cerebellar peduncle.

TMP13 p. 724

19. **E)** In this example, the flexor withdrawal reflex is activated by a painful stimulus to the right foot. Flexor muscles in the right leg and extensor muscles in the left leg are simultaneously stimulated to contract, causing reflex removal of the foot from the painful stimulus while shifting body weight to the other leg. The patellar tendon reflex (also called *knee jerk*), which is activated by tapping the patellar tendon, is a type of stretch reflex. The Golgi tendon reflex provides a negative feedback mechanism that prevents the development of too much tension in a muscle.

TMP13 pp. 702-704

20. **E)** The flexor withdrawal reflex is a polysynaptic reflex arc activated by stimulation of nociceptors in the skin. Multiple excitatory and inhibitory interneurons in the spinal cord are involved. The stretch reflex is a monosynaptic reflex arc involving two neurons. The Golgi tendon reflex is a disynaptic reflex arc because the reflex involves two synapses—an afferent and efferent neuron synapse with an inhibitory interneuron in the spinal cord.

TMP13 pp. 702-704

21. **A)** The hypothalamus, despite its small size, is the most important control center for the limbic system. It controls most of the vegetative and endocrine functions of the body and many aspects of behavior.

TMP13 p. 755

22. **C)** The association cortex is defined by the fact that it receives multiple inputs from a wide variety of sensory areas of cortex. It is the true multimodal cortex.

TMP13 pp. 739-740

23. **C)** The corpus callosum is the main fiber pathway for communication between the two hemispheres of the brain.

TMP13 pp. 741-742

24. **B)** Corticospinal fibers pass through the medullary pyramid.

TMP13 pp. 709-710

25. **D)** Prosopagnosia is the inability to recognize faces. This inability occurs in people who have extensive damage on the medial undersides of both occipital lobes and along the medioventral surfaces of the temporal lobes.

TMP13 p. 740

26. **E)** The somatic, visual, and auditory association areas all meet one another at the junction of the parietal, temporal, and occipital lobes. This area is known as *Wernicke's area*. This area on the dominant side of the brain plays the single greatest role for the highest comprehension levels we call intelligence.

TMP13 pp. 740-741

27. **B)** The combination of a motor neuron and all the muscle fibers innervated by that motor neuron is called a *motor unit*.

TMP13 p. 696

28. **E)** Broca's aphasia typically involves an inability to speak words correctly in the absence of any true paralysis of the laryngeal or pharyngeal musculature.

TMP13 pp. 708-709

29. **D)** A stroke involving the left middle cerebral artery is likely to cause an aphasic syndrome that may involve the loss of speech comprehension and/or the loss of the ability to produce speech sounds. Any paralysis resulting from the lesion would affect the right side of the body; similarly, any visual field deficits would affect the right visual field of each eye.

TMP13 p. 741

30. **E)** For an event or sensory experience to be remembered, it must first be consolidated. The consolidation of memory takes time. A disruption of consciousness during the process of consolidation will prevent the development of memory for the event or sensory experience.

TMP13 pp. 748-749

- 31. A)** The connection between Wernicke's area and Broca's area is made by the arcuate fasciculus.
TMP13 pp. 743-744
- 32. A)** Type 1a sensory fibers that innervate the stretch receptors of the muscle spindle travel in the appropriate spinal nerve that provides both the sensory and motor innervation of the muscle. The spinal nerves carry both afferent and efferent fibers. The afferent fibers (which contain the sensory fibers innervating the muscle spindle) pass through the dorsal root. Cutting the dorsal root will remove the afferent limb of the stretch reflex arc.
TMP13 pp. 696, 698-699
- 33. C)** Posterior spinocerebellar fibers pass through the inferior cerebellar peduncle.
TMP13 p. 723
- 34. D)** The main pathway linking the cerebral cortex and the cerebellum involves cortical projections to the ipsilateral basilar pontine nuclei, the cells of which then project to the contralateral cerebellum.
TMP13 pp. 722-723
- 35. C)** The cerebrospinal fluid outside the brain and spinal cord is located within the subarachnoid space. Dilated regions of the subarachnoid space are identified as cisterns. The cisterna magna is one of the largest cisterns and is positioned at the caudal end of the fourth ventricle between the cerebellum and posterior surface of the medulla.
TMP13 p. 790
- 36. D)** This woman has Huntington's disease. This hereditary disorder results from expansion of a CAG triplet repeat in the Huntingtin gene on chromosome 4. Typical symptoms are listed in the question stem. Huntington's disease is a neurodegenerative disorder that at first causes flicking movements in individual muscles and then progresses to distortional movements of the entire body; severe dementia develops along with the motor dysfunctions.
TMP13 pp. 734-735
- 37. C)** The abnormal movements of Huntington's disease are thought to be caused by loss of GABA-secreting neurons in the caudate nucleus and putamen; acetylcholine-secreting neurons in many parts of the brain are also thought to be affected. The axon terminals of GABA-secreting neurons normally inhibit portions of the globus pallidus and substantia nigra. This loss of inhibition is thought to allow spontaneous outbursts of globus pallidus and substantia nigra activity that cause the distortional movements.
TMP13 pp. 734-735
- 38. A)** Pontocerebellar axons are contained in the middle cerebellar peduncle.
TMP13 pp. 722-723
- 39. A)** Pain signals traveling through the anterolateral system, but not any of the discriminative sensations coursing through the medial lemniscal system, provide input to the cells in the reticular formation that give rise to ascending projections to the intralaminar nuclei of the thalamus.
TMP13 pp. 751-752
- 40. A)** Preganglionic sympathetic axons synapse on cells in the adrenal medulla that function as postganglionic sympathetic neurons.
TMP13 p. 774
- 41. D)** Gamma motor neurons innervate the contractile ends of the muscle spindle receptor. Stimulation of gamma motor neurons will cause the ends of the spindle to contract, which in turn will stretch the center of the spindle receptor in the muscle in which the spindle receptor is embedded. The activity of the gamma motor neurons is influenced by the fusimotor system. Enhanced activity of this system will lead to an increase in gamma motor tone and increase the sensitivity of the muscle spindle as a stretch receptor.
TMP13 pp. 699-670
- 42. D)** The cerebellum is responsible for coordinating and timing motor activity. Disorders of the cerebellum are associated with lack of coordination of motor activity. An example of this lack of coordination is ataxia, which is an unsteady gait.
TMP13 p. 729
- 43. E)** Preganglionic parasympathetic neurons that contribute to the innervation of the descending colon and rectum are found at S2 and S3 levels of the spinal cord.
TMP13 p. 775
- 44. A)** Complex spike output from the Purkinje cells of the cerebellum is a response to activation of climbing fibers in cerebellar neural circuitry. All climbing fibers originate in the inferior olivary nucleus.
TMP13 p. 724
- 45. D)** The dynamic nuclear bag fiber responds to the rate of change of length of the muscle spindle receptor. This fiber is responsible for the dynamic response of the muscle spindle.
TMP13 p. 698
- 46. A)** Cortical projections to the red nucleus provide an alternative pathway for the cerebral cortex to control flexor muscles through the rubrospinal tract.
TMP13 pp. 710-711
- 47. D)** The pontine reticular nuclei are tonically active. These nuclei have a stimulatory effect on the anti-gravity muscles of the body. The pontine nuclei are normally opposed by the medullary reticular nuclei. The medullary nuclei are not tonically active and require stimulation from higher brain centers to counterbalance the signal from the pontine nuclei. Decerebrate rigidity

results when the stimulatory signal from higher brain areas to the medullary nuclei is absent. This absence allows an unopposed and vigorous activation of the antigravity muscles, resulting in extension of the arms and legs and contraction of the axial muscles of the spinal column.

TMP13 p. 714

48. E) Corticospinal axons originate from cell bodies (pyramidal neurons) in layer V of the motor areas of the cortex.

TMP13 pp. 709-710

49. E) This man has Parkinson's disease. No laboratory biomarkers exist for Parkinson's disease, and imaging results are unremarkable. Diagnosis requires two of three cardinal signs that include (1) resting tremor, (2) rigidity, and (3) bradykinesia (or slow movement); this man has all three signs. Parkinson's disease affects about 1 percent of persons older than 60 years. Progressive disability can be slowed but not halted by treatment.

TMP13 p. 734

50. C) This man with Parkinson's disease has a loss of pigmented dopaminergic neurons of the substantia nigra pars compacta that send dopamine-secreting nerve fibers to the caudate nucleus and putamen. The causes of the abnormal motor movements are poorly understood; however, dopamine is an inhibitory transmitter in the caudate nucleus and putamen. It is therefore possible that overactivity of the caudate nucleus and putamen could result from decreased dopamine levels in this patient with Parkinson's disease; these brain structures are largely responsible for voluntary movement.

TMP13 p. 734

51. C) The palmar (volar) surfaces of the skin contain receptors that project through the medial lemniscal system to the primary somatosensory cortex. When these fingers are flexed and grasp an object, the cutaneous receptors send signals to the primary somatosensory cortex. These cortical neurons then project to the adjacent motor cortex and the pyramidal neurons that sent the original message down the corticospinal tract to cause contraction of the finger flexors. The motor cortex neurons are then said to be "informed of the muscle contractions" that they originally specified.

TMP13 pp. 711-712

52. B) Sweat glands and the piloerector smooth muscle of hairy skin are innervated by the population of cholinergic postganglionic sympathetic neurons.

TMP13 p. 775

53. D) Bilateral ablation of the amygdala causes behavioral changes known as *Klüver-Bucy syndrome*. These changes include lack of fear, extreme curiosity, forgetfulness, oral fixation, and a strong sex drive. The sex

drive can be so strong that monkeys will attempt to copulate with immature animals, animals of the wrong sex, and even animals of the wrong species. Although similar brain lesions in humans are rare, afflicted people have similar symptoms. The amygdala is thought to make the person's behavioral response appropriate for each occasion.

TMP13 p. 760

54. C) Although the majority of corticospinal axons synapse with the pool of spinal cord interneurons, some will synapse directly with the motor neurons that innervate muscles controlling the wrist and finger flexors.

TMP13 p. 712

55. A) The foramen of Magendie and the two lateral foramina of Luschka form the communication channels between the ventricular system within the brain and the subarachnoid space that lies outside the brain and spinal cord.

TMP13 pp. 790-791

56. A) A generalized tonic-clonic epileptic seizure is associated with the sudden onset of unconsciousness and an overall steady but uncoordinated contracture of many muscles of the body followed by alternating contractions of flexor and extensor muscles—that is, tonic-clonic activity. This effect is the result of widespread and uncontrolled activity in many parts of the brain. It takes the brain from a few minutes to a few hours to recover from this vigorous activity.

TMP13 pp. 769-770

57. B) Wernicke's area in the dominant hemisphere is responsible for interpreting spoken language. Damage to Wernicke's area will eliminate comprehension of spoken language.

TMP13 pp. 740-741

58. E) Afferent signals to the cerebellum travel primarily in the dorsal and ventral spinocerebellar tracts. The dorsal spinocerebellar tract carries signals from the muscle spindle receptors and Golgi tendon receptors, as well as large tactile receptors of the skin and joint proprioceptors. The ventral spinocerebellar tract carries information from the anterior portion of the spinal cord. This tract relays information regarding which motor signals from the motor areas of the brain have arrived at the level of the spinal cord.

TMP13 p. 723

59. D) Golgi tendon organs provide direct synaptic input to type Ib inhibitory interneurons. Type Ia interneurons and alpha motor neurons receive input from muscle spindle afferents, whereas dynamic gamma motor neurons and excitatory interneurons receive their input from supraspinal systems.

TMP13 p. 701

- 60. A)** Neurons in the locus coeruleus utilize the neurotransmitter norepinephrine in their widespread projections throughout the brain.
TMP13 pp. 752-753
- 61. D)** Dysdiadochokinesia is the inability to perform rapid alternating movements. Patients with hemineglect are unaware of items to one side of space. Astereognosis is the inability to recognize objects by touch. Agraphesthesia is a disorientation of the skin's sensation across its space (e.g., it is difficult to identify a number or letter traced on the hand). Dysarthria is a failure of progression in talking.
TMP13 pp. 729-730
- 62. A)** The cerebellum plays major roles in the timing of motor activities and in rapid, smooth progression from one muscle movement to the next. Lesions of the cerebellum can also cause dysmetria, ataxia, past pointing, nystagmus, dysarthria, intention tremor, and hypotonia. The premotor cortex and primary motor cortex plan and execute movements. The limbic system is involved with behavior, motivation, emotion, long-term memory, and olfaction.
TMP13 pp. 721-724
- 63. A)** The excitatory or inhibitory effect of a postganglionic sympathetic fiber is determined solely by the type of receptor to which it binds.
TMP13 p. 777
- 64. A)** CSF will flow across the valve-like arachnoid villi when the CSF pressure is only a few millimeters higher than the pressure within the superior sagittal sinus.
TMP13 p. 792
- 65. C)** The most characteristic deficit after damage to corticospinal tract neurons involves discrete voluntary movement of the contralateral hand and fingers.
TMP13 p. 713
- 66. A)** A large area of the primary motor cortex is dedicated to activating the muscles that control the movement of the fingers. Stimulation of the primary motor cortex usually results in very discrete contractions of small groups of muscles. Stimulation of the premotor cortex results in the contraction of large groups of muscles, and stimulation of the supplemental motor area results in bilateral movements.
TMP13 pp. 707-708
- 67. B)** Nicotinic cholinergic receptors are found at synapses between preganglionic and postganglionic sympathetic neurons.
TMP13 p. 777
- 68. D)** The premotor cortex generates nerve signals for complex patterns of movement rather than discrete patterns generated in the primary motor cortex. The most anterior part of the premotor area first develops a motor image of the total muscle movement that is to be performed. Next, the successive pattern of muscle activity required to achieve the image excites neurons in the posterior premotor cortex; from here, signals are sent directly to the primary motor cortex to excite specific muscles or by way of the basal ganglia and thalamus and then to the primary motor cortex.
TMP13 p. 701
- 69. C)** The perivascular space (also known as the *Virchow-Robin space*) is formed between the outer wall of small vessels penetrating into the brain and the pia mater, which lines the outer surface of the brain and is only loosely attached to the brain.
TMP13 p. 792
- 70. D)** The spike and dome pattern is characteristic of an absence seizure.
TMP13 p. 770
- 71. D)** The concentration of protein in the CSF is only 1 percent to 2 percent of that of plasma; interstitial fluid in the tissues of the brain has an equally low protein concentration. This low protein concentration of CSF can be attributed to the blood-brain barrier, which is impermeable to protein. The concentration of sodium in CSF is slightly less than that of plasma, the chloride concentration is about 15 percent greater than that of plasma, the potassium concentration is 40 percent of that of plasma, and the glucose concentration of CSF is about 30 percent of that of plasma.
TMP13 p. 791
- 72. A)** Active transport of sodium ions through the epithelial cells lining the choroid plexus is followed by passive diffusion of chloride ions to maintain electroneutrality. The osmotic gradient created by the sodium and chloride ions causes the immediate osmosis of water into the CSF. The osmolarity of CSF is identical to that of blood plasma.
TMP13 pp. 790-791
- 73. A)** This patient has Friedreich's ataxia, which is an autosomal-recessive ataxia resulting from a mutation on chromosome 9. It accounts for about 50 percent of all hereditary ataxias. Huntington's disease is a neurodegenerative disease that affects muscle coordination and causes a decline in cognitive function and psychiatric problems. Multiple sclerosis is an inflammatory disease in which the myelin covering of nerve cells in the brain and spinal cord is damaged, resulting in a wide range of symptoms that include physical, mental, and psychiatric problems.
TMP13 pp. 729-730
- 74. E)** The major pathological finding in Friedreich's ataxia is degeneration and loss of axons, especially in the spinal cord and spinal roots; this effect increases with age and duration of disease. Most major nerve tracts in the spinal cord show demyelination, and the spinal cord itself

becomes thin. There are no lesions in the premotor cortex or primary motor cortex, and the frontal lobe remains normal. The disorder does not affect cognitive functions, and unmyelinated sensory fibers are spared.

TMP13 pp. 729-730

75. **B)** Cells in the pars compacta portion of the substantia nigra use the neurotransmitter dopamine in their projections to the caudate and putamen.

TMP13 p. 733

76. **A)** Lesions that damage primary motor cortex and other surrounding motor cortical areas lead to spastic paralysis in the affected muscles.

TMP13 p. 713

77. **D)** The parahippocampal gyrus is an important component of the limbic cortex, or limbic lobe.

TMP13 pp. 754-755

78. **C)** Epinephrine activates alpha- and beta-adrenergic receptors equally well. Norepinephrine excites both types of receptors but has a markedly greater effect on alpha receptors.

TMP13 p. 777

79. **A)** The posterior and lateral hypothalamus, in combination with the preoptic hypothalamus, form an important group of cells controlling cardiovascular functions such as heart rate and blood pressure.

TMP13 pp. 755-756

80. **A)** Ia sensory fibers synapse directly with alpha motor neurons, whereas Ib sensory fibers synapse with inhibitory interneurons. Excitatory interneurons play an important role in the withdrawal reflex. Gamma motor neurons receive input primarily from supraspinal systems.

TMP13 pp. 698-700

81. **E)** Noncommunicating hydrocephalus results when a blockage of CSF flow occurs within the ventricular system or at the sites of communication between the ventricular system and the subarachnoid space. Communicating hydrocephalus occurs when a blockage occurs either within the subarachnoid space or at the arachnoid villi, thus preventing communication between the subarachnoid space and the superior sagittal sinus.

TMP13 p. 793

82. **E)** Behavioral deficits, changes in personality, and diminished problem-solving ability are all signs of damage to the prefrontal association cortex.

TMP13 pp. 741-742

83. **A)** Wernicke's cortical area is the major brain area for language comprehension. A person with a lesion in Wernicke's area may be able to understand either the spoken word or the written word but would not be able to interpret the thought that is expressed.

TMP13 pp. 739-740

84. **C)** The tight junctions formed between adjacent choroid epithelial cells represent the structural basis of the blood-CSF barrier. The blood-brain barrier is formed by the tight junctions between adjacent endothelial cells of brain capillaries.

TMP13 pp. 793-794

85. **C)** The withdrawal reflex is activated by stimuli from free nerve endings. Muscle spindles provide the afferent signals for the stretch reflex, and Golgi tendon organs are the source of stimuli for the inverse myotatic reflex.

TMP13 pp. 701-702

86. **B)** The posterior temporal lobe is larger at birth in the dominant hemisphere of the brain, which is the left hemisphere in 95 percent of people. Because of the tendency to direct one's attention to the better developed region, the rate of learning in the cerebral hemisphere that gains the first start increases rapidly, whereas learning remains slight in the opposite, less-used side. Hence, the left hemisphere normally becomes dominant over the right.

TMP13 pp. 741-742

87. **A)** The nasal, lacrimal, salivary, and gastrointestinal glands are stimulated by cholinergic postganglionic parasympathetic neurons.

TMP13 pp. 778, 780

88. **A)** The neurons located in the locus coeruleus release norepinephrine at their nerve terminals.

TMP13 pp. 752-753

89. **A)** The crossed extensor reflex depends on incoming pain signals distributed to both sides of the spinal cord via excitatory interneurons.

TMP13 p. 703

90. **D)** The cerebocerebellum and the dentate nucleus are involved with the thalamus and cortex in the planning of complex movements.

TMP13 p. 722

91. **D)** The stretch reflex is mediated by muscle spindles. Autogenic inhibition involves Golgi tendon organs. Reciprocal inhibition is also related to muscle spindles.

TMP13 pp. 698-699

92. **C)** The junction of the parietal, temporal, and occipital lobe is commonly referred to as *Wernicke's area*. This area of the brain is responsible for the ability to comprehend both the written and spoken word.

TMP13 p. 739

93. **D)** The corpus callosum connects the left and right cerebral hemispheres and hence facilitates communication between them. Agenesis of the corpus callosum is a rare defect in which there is a complete or partial absence of the corpus callosum.

TMP13 pp. 709-710, 742

- 94. D)** Damage to the subthalamic nucleus of the basal ganglia often leads to flailing movements of an entire limb; this condition is called *hemiballismus*. Stroke is the most common cause of hemiballismus in adults, but this condition is rare. The globus pallidus is part of the basal ganglia and is involved with movement; however, damage to the globus pallidus does not cause hemiballismus. The lateral hypothalamus is mostly concerned with hunger. The red nucleus serves as an alternative pathway for transmitting cortical signals to the spinal cord; it controls the crawling of babies and may be responsible for swinging the arms while walking. The ventrobasal complex of thalamus is a sensory relay area of the brain.
TMP13 p. 732
- 95. F)** The greater reduction in blood flow to the front limbs is caused by denervation supersensitivity of norepinephrine receptors. The mechanism of denervation supersensitivity is poorly understood, but it is most likely caused by an actual increase in the number of norepinephrine receptors on the muscle vasculature. These additional receptors greatly enhance the vasoconstrictor effects of norepinephrine.
TMP13 p. 777
- 96. B)** The amygdala seems to function in behavioral awareness at a semiconscious level. The amygdala also is thought to project into the limbic system the individual's current status with respect to his or her surroundings. Therefore, the amygdala is believed to help pattern behavior appropriate for each occasion.
TMP13 p. 760
- 97. A)** The innervation and function of systemic blood vessels is influenced primarily, if not exclusively, by the sympathetic nervous system.
TMP13 p. 780
- 98. E)** Schizophrenia is thought to be caused in part by excessive release of dopamine. Occasionally, patients with Parkinson's disease exhibit schizophrenic symptoms because of uncontrolled L-dopa therapy and the subsequent production of dopamine.
TMP13 p. 771
- 99. C)** The magnocellular portion of the red nucleus has a somatographic representation of all the muscles of the body, similar to the motor cortex. Stimulation of this area in the red nucleus results in contraction of a single muscle or small groups of muscles.
TMP13 p. 711
- 100. C)** Lesions involving the ventromedial hypothalamus lead to excessive eating (hyperphagia), excessive drinking, rage and aggression, and hyperactivity.
TMP13 p. 757
- 101. B)** The high metabolic rate in the nervous system is primarily due to the high metabolic activity in neurons, even in the resting state.
TMP13 p. 794
- 102. D)** The vermis and the intermediate zone of the cerebellar hemisphere have a distinct topographic representation of the body. These areas are responsible for coordinating the contraction of the muscles of the body for intended motion.
TMP13 p. 722
- 103. E)** The medial forebrain bundle extends from the septal and orbitofrontal regions of the cerebral cortex downward through the center of the hypothalamus to the brain stem reticular area. This structure serves as an important communication system between the limbic system and the brain stem.
TMP13 p. 755
- 104. C)** This woman has Alzheimer's disease. Increased amounts of beta-amyloid peptide is found in the brains of patients with Alzheimer's disease. The peptide accumulates in amyloid plaques with diameters up to several hundred millimeters in widespread areas of the brain, including the cerebral cortex, hippocampus, basal ganglia, thalamus, and cerebellum. A key role for excess accumulation of beta-amyloid peptide in the pathogenesis of Alzheimer's disease is suggested by multiple observations.
TMP13 pp. 771-772
- 105. E)** Dysdiadochokinesia is a cerebellar deficit that involves a failure of progression from one part of a movement to the next. Consequently, movements that include rapid alternation between flexion and extension are most severely affected.
TMP13 p. 729
- 106. C)** Linear acceleration is in a straight line; angular acceleration is that which occurs by turning about a point. The semicircular canals respond to the turning motions of the head and body.
TMP13 p. 717
- 107. D)** Decreased activity of postganglionic sympathetic neurons leads to vasodilation of systemic arterioles. In contrast, increased activity in postganglionic sympathetics results in vasoconstriction.
TMP13 p. 782
- 108. B)** Damage to Broca's area leads to motor aphasia, or the inability to form words correctly.
TMP13 p. 744
- 109. D)** This man has schizophrenia, which is characterized by a breakdown of cognitive and emotional responses. Dissociative identity disorder was formerly called *multiple personality disorder*. Bipolar disorder

is characterized by episodes of elevated mood (mania) alternating with episodes of depression.

TMP13 p. 771

- 110. B)** A consistent finding in most schizophrenics is that the hippocampus is reduced in size. The hippocampus is part of the limbic system. Incoming sensory information activates various parts of the hippocampus that, in turn, initiate behavioral reactions for different purposes. Removal of the hippocampus makes it impossible to learn new information based on verbal symbolism; however, past memories are preserved.

TMP13 p. 771

- 111. B)** Hair cells in the macula of the saccule are maximally sensitive to linear head movement in the vertical plane.

TMP13 p. 716

- 112. D)** Lesions involving the thalamus lead to retrograde amnesia, because they are believed to interfere with the process of retrieving long-term memory stored in other portions of the brain.

TMP13 p. 749

- 113. C)** The caudate nucleus is involved in the basal ganglia circuits that control memory-guided motor activity.

TMP13 pp. 732-733

- 114. C)** This infant has shaken baby syndrome. The subdural hematoma has increased intracranial pressure, which in turn has caused cerebral edema. The venous vasculature in the brain is compressed due to the high intracranial pressure. Continued compression of brain structures can lead to worsening cerebral edema with decreased oxygenation of the brain.

TMP13 p. 793

- 115. B)** The punishment center is primarily localized to the periventricular hypothalamus and the midbrain central gray area.

TMP13 p. 758

- 116. A)** Athetosis is a slow and continuous writhing movement of the arm, neck, or the face. It results from damage or dysfunction of the globus pallidus.

TMP13 p. 731

- 117. A)** An example of a relatively restricted or local sympathetic action is the vasodilation or vasoconstriction of blood vessels that occurs upon warming or cooling of a patch of skin. When a bright light is introduced to one eye, the pupils of both eyes constrict. The pupillary light reflex is a multiple-neuron event that involves the Edinger-Westphal nucleus of the brainstem; it is not a local event.

TMP13 p. 782

- 118. E)** The most conspicuous stimulation area for causing sleep is the raphe nuclei in the lower half of the pons and in the medulla. Many nerve endings of fibers from raphe neurons secrete serotonin. When the formation of serotonin is blocked by drugs, sleep is often disrupted for hours to days. Therefore, it has been assumed that serotonin is a transmitter associated with the production of sleep.

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- 119. D)** The Golgi tendon organ senses tension in the tendons. When tension becomes exceedingly high, an inhibitory reflex is activated that causes relaxation of the entire muscle, which serves to protect the muscle from tearing. However, the Golgi tendon organ is also thought to play a key role in maintaining equal tension in the muscle fibers of a skeletal muscle so that imbalances in tension among the different muscle fibers can be equalized.

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- 120. B)** The fornix connects the hippocampus to the anterior thalamus, hypothalamus, and the limbic system.

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- 121. B)** Degeneration of the dopaminergic cells in the pars compacta of the substantia nigra is thought to be the primary defect in Parkinson's disease.

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- 122. E)** Phenylephrine is a sympathomimetic drug that stimulates adrenergic receptors. Reserpine, phentolamine, and propranolol are sympathetic antagonists.

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